Dr. A.P.J. ABDUL KALAM TECHNICAL UNIVERSITY

LUCKNOW, UTTAR PRADESH

EVALUATION SCHEME & SYLLABUS

of

B. TECH.

First Year

BT STREAM

(*Biotechnology, Biomedical Engineering, Bioinformatics*)

AS PER AICTE MODEL CURRICULUM & NEP2020

(Effective from the Academic Session: 2026-27)

BT Stream

Dr. A.P.J. Abdul Kalam Technical University, Lucknow- Uttar Pradesh

Bachelor of Technology (BT Stream)

B.TECH. FIRST YEAR, SEMESTER-I

Scheme of Teaching and Examinations (w.e.f. 2026-27)

I Semester

3- WEEKS STUDENT INDUCTION PROGRAMME IN THE BEGINNING OF THE SESSION AS PER AICTE GUIDELINES

Sr. No.	Course Category	Course Code	Course Title	Teaching Department	L	T	P	NCrF Hrs.	IA (TS/PS) (CT+TA)	EA (TE/PE)	Total Marks (IA+EA)	Credits
1	BSC	AAS101F/ AAS102F	Physics of Materials and Biomaterials / Applied Chemistry for Biomolecules and Bioprocess Systems	AS&H	3	0	0	45	30	70	100	03
2	BSC	AAS103F	Calculus and Linear Algebra	AS&H	3	1	0	60	30	70	100	04
3	ESC	AEE101C/ AEC101C	Fundamentals of Electrical Engineering / Fundamentals of Electronics Engineering	EE/ ECE	2	1	0	45	30	70	100	03
4	ESC	ACS101	Programming Languages	CSE	2	1	0	45	30	70	100	03
5	HSMC	AAS104/ AAS105	Introduction to Indian Knowledge System/ Professional Communication & Technical Writing	AS&H	2	0	0	30	30	70	100	02
6	ESC	AAI101/ AME101C	Introduction to AI & Prompt Engineering/ Fundamentals of Mechanical Engineering	Respective Deptt./ ME	2	0	0	30	30	70	100	02
7	BSC	AAS151/ AAS152F	Applied Physics Lab/ Applied Chemistry for Biomolecules and	AS&H	0	0	2	30	50	50	100	01

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			Bioprocess Systems Lab									
8	ESC	AEE151C/ AEC151C	Electrical Engineering Lab / Electronics Engineering Lab	EE /ECE	0	0	2	30	50	50	100	01
9	ESC/ SDC	ACE151G/ AID151	Computer-Aided Engineering Drawing Lab/Innovation & Design Thinking Lab (Project-Based Learning)	CE/ Respective Engg.	0	1	2	45	50	50	100	02
10	BSC/ NCMC	AAS153/ AAC101	Language Lab/ Environment & Sustainability	AS&H	0	0	2	30	50	50	100	-
					2	0	0	30	30	70	100	-
11	NCMC	AGP151	General Proficiency	Respective Deptt.	0	0	2	30	50	—	50	-
				Total	**16**	**04**	**10**	**450**	**330**	**570**	**900**	**21**

AICTE Activity Points (Mandatory)
(i) Students must earn **100** activity points between Semester 1 to 8
(ii) Mandatory requirement for award of degree

Course Category Abbreviations:
BSC *– Basic Science Course;* ***ESC*** *– Engineering Science Course;* ***PSC*** *– Programme Specific Course;* ***AEC*** *– Ability Enhancement Course;* ***NCMC*** *– Non-Credit Mandatory Course;* ***SDC*** *– Skill Development Course;* ***HSMC*** *– Humanities, Social Science & Management Course;* ***AS&H****- Applied Science and Humanities*

BT Stream

UG Programmes Stream-wise Classification

Sr. No.	Stream	Branches
1	**Civil Engineering Stream (CE)**	Civil Engineering, Environmental Engineering
2	**Mechanical Engineering Stream (ME)**	Mechanical Engineering, Automobile Engineering, Aeronautical Engineering, Industrial Production Engineering, Manufacturing Technology, Mining Engineering
3	**Electrical Engineering Stream (EE)**	Electrical Engineering, Electrical & Electronics Engineering, Electrical & Computer Engineering
4	**Electronics Engineering Stream (ECE)**	Electronics Engineering, Electronics and Communication Engineering, Electronics and Communication (Advanced Communication Technology), Electronics Engineering (VLSI Design and Technology), Electronics and Instrumentation Engineering, Instrumentation and Control Engineering, Electronics and Computer Engineering
5	**Computer Science & Engineering Stream (CSE)**	Computer Science and Engineering, Computer Science, Computer Engineering, Information Technology, Computer Science and Information Technology, Computer Science & Technology, Computer Science and Design, Computer Science and Engineering (Hindi), Computer Science and Engineering (Artificial Intelligence), Computer Science and Engineering (Artificial Intelligence & Machine Learning), Artificial Intelligence and Machine Learning, Artificial Intelligence (AI) and Data Science, Computer Science and Engineering (Data Science), Computer Science & Engineering (Cyber Security), Computer Science and Engineering (Internet of Things), Cyber Security, Data Science
6	**Textile Engineering Stream (TX)**	Textile Technology, Handloom & Textile Technology, Carpet & Textile Technology
7	**Biotechnology Stream (BT)**	Biotechnology, Biomedical Engineering, Bioinformatics
8	**Chemical Engineering (CH)**	Chemical Engineering, Plastic Engineering, Food Technology
9	**Agricultural Engineering (AG)**	Agricultural Engineering

BT Stream

Dr. A.P.J. Abdul Kalam Technical University, Lucknow- Uttar Pradesh

Bachelor of Technology (BT Stream)

B.TECH. FIRST YEAR, SEMESTER-II

Scheme of Teaching and Examinations (w.e.f. 2026-27)

II Semester												
Sr. No.	**Course Category**	**Course Code**	**Course Title**	**Teaching Department**	**L**	**T**	**P**	**NCrF Hrs.**	**IA (TS/PS) (CT+TA)**	**EA (TE/PE)**	**Total Marks (IA+EA)**	**Credits**
1	BSC	AAS201F/ AAS202F	Physics of Materials and Biomaterials / Applied Chemistry for Biomolecules and Bioprocess Systems	AS&H	3	0	0	45	30	70	100	03
2	BSC	AAS203F	Calculus and Numerical Methods	AS&H	3	1	0	60	30	70	100	04
3	ESC	AEE201C/ AEC201C	Fundamentals of Electrical Engineering / Fundamentals of Electronics Engineering	Respective Engg. Deptt.	2	1	0	45	30	70	100	03
4	ESC	ACS201	Essentials of Data Structure	CSE	2	1	0	45	30	70	100	03
5	HSMC	AAS204/ AAS205	Introduction to Indian Knowledge System /Professional Communication & Technical Writing	AS&H	2	0	0	30	30	70	100	02
6	ESC	AAI201/ AME201C	Introduction to AI & Prompt Engineering/ Fundamentals of	AS&H/ Respective Engg. Deptt.	2	0	0	30	30	70	100	02

BT Stream

			Mechanical Engineering									
7	BSC	AAS251/ AAS252F	Applied Physics Lab/ Applied Chemistry for Biomolecules and Bioprocess Systems Lab	AS&H	0	0	2	30	50	50	100	01
8	ESC	AEE251C/ AEC251C	Electrical Engineering Lab /Electronics Engineering Lab	EE Deptt. / ECE Deptt.	0	0	2	30	50	50	100	01
9	ESC/ SDC	ACE251G/ AID251	Computer-Aided Engineering Drawing Lab/Innovation & Design Thinking Lab (Project-Based Learning)	CE Deptt./ Respective Engg. Deptt.	0	1	2	45	50	50	100	02
10	BSC/ NCMC	AAS253/ AAC201	Language Lab /Environment & Sustainability	AS&H	0	0	2	30	50	50	100	-
					2	0	0	30	30	70	100	-
11	NCMC	AGP251	General Proficiency	Respective Deptt.	0	0	2	30	50	—	50	-
				Total	**16**	**04**	**10**	**450**	**330**	**570**	**900**	**21**

The **Student Induction Programme** (SIP), initiated by the All India Council for Technical Education (AICTE), is designed to help newly admitted students in technical institutions transition smoothly into the higher education environment. It aims to familiarize students with the institutional culture, foster connections with peers and faculty, and provide a foundation for holistic learning. Activities under SIP may include Physical Activities, Creative Arts, Universal Human Values, Literary Events, Proficiency Modules. Lectures shall be by Eminent Personalities, Local Area Visits, Department/Branch Familiarization, and Innovation-related sessions.

AICTE Activity Points Requirement for B.Tech. Programmes

As per AICTE guidelines (refer Chapter 6 – *AICTE Activity Point Program, Model Internship Guidelines*), in addition to academic requirements, students must earn a specified number of Activity Points to be earned is to be eligible for the award of their degree.

- **Regular** students admitted to a 4-year degree program must earn **100 Activity Points**.
- **Lateral** entry students (joining from the second year) must earn **75 Activity Points**.
- **Students** transferred from other universities directly into the fifth semester must earn **50 Activity Points** from the date of entry into AKTU.

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These Activity Points are **non-credit** and will not be considered for **the SGPA/CGPA** or be used for **vertical progression**. However, they are mandatory for the **award of the degree**, and the points earned will be reflected on the **eighth semester Grade Card**.
The hours spent for earning the activity points shall not be counted for regular attendance requirements. Students can accumulate these points at any time during their program, including on weekends, holidays, and vacations starting from the year of admission, provided they meet the minimum hours of engagement prescribed for each activity. If a student fails to earn the required Activity Points, the eighth-semester Grade Card will be withheld until the requirement is fulfilled. Consequently, the degree will be awarded only after the Grade Card has been released

BT Stream

Course Code	**AAS101F/AAS201F**
Title of the Course	Physics of Materials and Biomaterials
L-T-P	3-0-0
Total Credits of the Course	03
NCrF Hours	45

Main Focus of the Course Outcomes

Employability	**Skill Development**	**Entrepreneurship**
✓	✓	✓

Course Objectives:

1. To understand the foundational concepts and principles of quantum mechanics.
2. To understand the fundamental characteristics of solid-state materials and to apply this knowledge to solve real-world problems in science and engineering.
3. To understand the dielectric and magnetic materials and their roles in various technological devices and systems.
4. To comprehend the principles and applications of photonics.
5. To explore the fundamental properties and phenomena of superconductivity and diverse applications of nano-materials, in advanced technological contexts.

Course Content

Unit	**Description**	**Weightage**
1.	**Fundamentals of Quantum Mechanics** Inadequacy of classical mechanics, de-Broglie concept of matter waves, Davisson and Germer experiment, wave packet, phase velocity, group velocity, Heisenberg's uncertainty principle and its applications, time-dependent and time-independent Schrodinger wave equations, particle confined in one dimensional box, wave function and its physical interpretation.	20%
2.	**Physical Properties of Materials and Surface Physics** Atomic structure and bonding (ionic, covalent, metallic), types of crystal structures (BCC, FCC, HCP), lattice parameters, amorphous structures, mechanical	20%

Draft Syllabus

	properties (stress, strain, elasticity, viscoelasticity), surface energy, wettability, contact angle, adhesion and cohesion.	
3.	**Dielectric and Magnetic Materials** **Dielectric materials:** Dielectrics, dielectric constant, electric displacement and polarization vectors, electric susceptibility, polarization mechanism, applications of dielectric materials. **Magnetic Materials**: Magnetic susceptibility, dia-, para-, ferro- materials, hysteresis loss, coercivity, remanence, importance of hysteresis curve, applications of magnetic materials.	20%
4.	**Photonics** **Fiber optics:** Fundamental idea about optical fiber, acceptance angle, acceptance cone, numerical aperture, principle of light propagation in optical fiber, attenuation and attenuation coefficient in fibre. **Laser:** Interaction of radiation with matter (absorption, spontaneous and stimulated emission), principle of laser operation, Einstein's coefficients, Ruby laser, He-Ne laser, laser applications.	20%
5.	**Superconductivity and Nano-materials** **Superconductors:** Temperature dependence of resistivity and critical field in superconducting materials, Meissner effect, Type I and Type II superconductors, persistent current, high temperature superconductors, properties, applications in biotechnology. **Nano-materials:** Introduction and properties of nano-materials, basic concept of quantum dots, quantum wires and quantum well, application of nano-materials.	20%

BT Stream

Teaching-Learning Methodology

Topics are taught through interactive lectures using blackboard and ICT tools. Laboratory demonstrations, case studies and discussions are incorporated. Mini-projects design may be included.

Course Outcomes (COs)

COs	Course Outcome Statement	KL
CO1	Understand the foundational concepts and principles of quantum mechanics.	K2
CO2	Understand the fundamental characteristics of solid-	K2

	state materials and to apply this knowledge to solve real-world problems in science and engineering.	
CO3	Understand the dielectric and magnetic materials and their roles in various technological devices and systems.	K2
CO4	Apply the principles and applications of photonics.	K3
CO5	Explore the fundamental properties and phenomena of superconductivity and diverse applications of nano-materials, in advanced technological contexts.	K2

Course Outcomes (COs)

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	–	1	1	–	–	–	–	–	–	1
CO2	3	2	1	2	2	–	–	–	–	–	–	1
CO3	3	3	2	2	2	1	1	–	–	–	–	1
CO4	3	2	2	2	3	1	1	–	–	–	–	1
CO5	3	2	2	2	2	1	2	–	–	–	–	2

Evaluation Pattern

Sr. No.	Details of the Evaluation	Weightage
1.	Internal Assessment (CT + Assignment + Quiz + Attendance)	30%
2.	External Assessment	70%

Suggested Textbooks / Reference Books

1. Engineering Physics: Theory and Practical- Katiyar and Pandey (Wiley India)
2. Biophysics – Roland Glaser (Springer)
3. Concepts of Modern Physics - Aurthur Beiser (Mc-Graw Hill)
4. Introduction to Quantum Mechanics – David J. Griffiths (Prentice Hall)
5. Engineering Physics Theory and Experiments-Srivastava and Yadav (New Age International)
6. Engineering Physics - S K Gupta (Krishna Prakashan)

Online Resources

1. NPTEL.
2. Coursera – Bioinformatics and Biomaterials specializations.
3. NIH and WHO digital resources.

Back to Course Structure & Evaluation Scheme

BT Stream

Course Code	**AAS102F/ AAS202F**
Title of the Course	Applied Chemistry for Biomolecules and Bioprocess Systems
Total Credits of the Course	03
NCrF Hours	45

Main Focus of the Course Outcomes

Employability	**Skill Development**	**Entrepreneurship**
✓	✓	✓

Course Objectives:

1. To understand the chemical structure and function of biomolecules including carbohydrates, proteins, lipids, and nucleic acids
2. To apply thermodynamic and kinetic principles to analyze biochemical reactions and metabolic pathways
3. To study the chemistry of enzymes, enzyme kinetics, and their applications in bioprocessing
4. To analyze the chemistry of fermentation media, bioreactor systems, and downstream processing
5. To introduce the chemistry of biopharmaceuticals, biosensors, and green biotechnology applications

Course Content

Unit	**Description**	**Weightage**
1.	**Chemistry of Biomolecules:** Structure and classification of carbohydrates (mono-, di-, polysaccharides), proteins (primary to quaternary structure, amino acid chemistry), lipids (fatty acids, phospholipids, sterols), nucleic acids (DNA and RNA structure, nucleotide chemistry), vitamins and coenzymes, biomolecular interactions and forces (hydrogen bonds, hydrophobic interactions, Van der Waals)	20%
2.	**Biochemical Thermodynamics and Kinetics:** Laws of thermodynamics applied to biological	20%

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	systems, Gibbs free energy, ATP as energy currency, coupled reactions, chemical equilibrium in biochemical systems, reaction kinetics (first and second order), activation energy, Arrhenius equation, transition state theory, introduction to enzyme catalysis and factors affecting enzyme activity.	
3.	**Enzyme Chemistry and Bioprocess Applications:** Classification and nomenclature of enzymes, active site chemistry, mechanisms of enzyme action, Michaelis-Menten kinetics, Km and Vmax, enzyme inhibition (competitive, non-competitive, uncompetitive), allosteric regulation, immobilized enzymes, industrial enzymes (proteases, amylases, lipases, cellulases), enzyme engineering	20%
4.	**Fermentation Chemistry and Bioprocessing:** Introduction to fermentation technology,Chemistry of fermentation media (carbon sources, nitrogen sources, minerals, growth factors), sterilization chemistry, aerobic and anaerobic fermentation, metabolic pathways (glycolysis, TCA cycle, pentose phosphate), yield coefficients, downstream processing (precipitation, chromatography, membrane filtration), product recovery	20%
5.	**Biopharmaceuticals and Green Biotechnology:** Chemistry of monoclonal antibodies, recombinant proteins, vaccines, nucleic acid therapeutics (siRNA, mRNA), biosimilars, drug formulation chemistry, biosensor chemistry (enzyme biosensors, immunosensors), bioremediation chemistry, Synthesis of typical organic compounds by conventional and Green route (Paracetamol), Environmental impact of Green chemistry on society.	20%

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Teaching-Learning Methodology

Topics are taught through interactive lectures using blackboard and ICT tools. Laboratory demonstrations on enzyme kinetics, fermentation, and biomolecule characterization are included. Case studies on

biopharmaceutical manufacturing and green bioprocessing supplement theoretical content.

Course Outcomes (COs)

COs	Course Outcome Statement	KL
CO1	Understand the chemical structure and functional significance of biomolecules	K2
CO2	Apply thermodynamic and kinetic principles to analyze biochemical reactions and metabolic processes	K3
CO3	Understand enzyme kinetics and apply enzyme chemistry to industrial bioprocess design	K3
CO4	Understand fermentation chemistry and apply it to bioprocess media design and product recovery	K3
CO5	Apply chemistry of biopharmaceuticals, biosensors, and green biotechnology to real-world applications	K3

CO-PO Articulation Matrix

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	1	1	1	–	–	–	–	–	–	1
CO2	3	3	2	2	2	–	–	–	–	–	–	1
CO3	3	3	3	2	2	1	1	–	–	–	–	1
CO4	3	2	3	2	2	1	1	–	–	–	–	1
CO5	3	2	2	2	2	2	3	2	–	–	–	2

Evaluation Pattern

Sr. No.	Details of the Evaluation	Weightage
1.	Internal Assessment (CT + Assignment + Quiz + Attendance)	30%
2.	External Assessment	70%

Suggested Textbooks / Reference Books

1. Engineering Chemistry – Jain and Jain
2. Biochemistry – Lehninger, Nelson, and Cox
3. Bioprocess Engineering Principles – Pauline M. Doran
4. Enzyme Biochemistry – T. Palmer
5. Industrial Biotechnology – Wittrup and Verdine (Eds.)

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Online Resources

1. NPTEL courses on Biochemistry and Bioprocess Engineering
2. Coursera – Biochemistry specializations (Duke, Johns Hopkins)
3. NCBI and UniProt online biochemical databases and resources

Additional References

Relevant review articles, research papers, and handouts on latest developments in:

1. mRNA Therapeutics: Chemistry and Formulation
2. Metabolic Engineering for Bio-based Chemical Production
3. Bioremediation: Chemical and Microbiological Approaches

Back to Course Structure & Evaluation Scheme

BT Stream

Course Code	**AAS103F**
Title of the Course	Calculus and Linear Algebra
Total Credits of the Course	04
NCrF Hours	60

Main Focus of the Course Outcomes

Employability	**Skill Development**	**Entrepreneurship**
✓	✓	✓

Course Objectives:

1. Apply differential calculus to solve biotechnology problems involving drug delivery, diffusion, and pharmacokinetics.
2. Use integral calculus to analyze enzyme kinetics, drug concentration, diffusion, and biological systems.
3. Formulate and solve differential equations for biotechnology applications such as population growth and bioreactor modeling.
4. Apply matrix algebra to solve linear systems in bioreactor, metabolic, and transport models.
5. Use vector calculus to analyze biological systems, biomechanics, biomedical imaging, and transport phenomena.

Course Content

Unit	**Description**	**Weightage**
1.	**Differential Calculus:** Functions, Limit, Continuity and Differentiability. Definition of derivative, physical and geometrical significance of derivative, derivative by first principal. Derivative of sum, difference, product and quotient of functions. Derivatives of polynomial and trigonometric functions, derivative of composite functions, chain rule, derivatives of inverse trigonometric functions, derivative of implicit function. Concept of exponential and logarithmic functions and their derivative. Logarithmic differentiation. Derivative of functions expressed in parametric forms. Second order derivatives. Applications of Derivatives: rate of change, maxima and minima (second derivative	20%

BT Stream

	test only). Simple problems (that illustrate basic principles and understanding of the subject as well as real life situations). **Applications:** Drug delivery and Pharmacokinetics, diffusion and mass transfer.	
2.	**Integral Calculus:** Integration as inverse process of differentiation. Integration of a variety of functions by substitution, by partial fractions and by parts, only simple integrals of the type to be evaluated. Basic properties of definite integrals and evaluation of definite integrals. Applications of the Integrals: Applications in finding the area under simple curves, especially lines, areas of circles/parabolas/ellipses (in standard form only). **Applications**: Bioreactor engineering, cell growth and biomass estimation, calculation of total drug exposure (Area Under the Curve, AUC), estimation of cumulative drug release.	20%
3.	**Differential Equations:** Definition, order and degree, general and particular solutions of a differential equation. Formation of differential equation whose general solution is given. Solution of differential equations of first order and first degree by method of separation of variables, homogeneous differential equations. Solutions of linear differential equation of the type: $\frac{dy}{dx} + py = q$, where p and q are functions of x. **Applications:** Population growth, Pharmacokinetics, diffusion and bioreactor modelling.	20%
4.	**Introduction to Matrices:** Definition, Types of Matrices, Addition, Subtraction, Scalar Multiplication and Multiplication of Matrices, Determinants: Definition, Minors, Cofactors, Properties of Determinants. Adjoint, Inverse and solution of system of linear equations. **Applications**: Solve system of linear equations in bioreactor and metabolic models, diffusion and transport problems.	20%
5.	**Vector Analysis**: Introduction to Vector Algebra: Scalar (dot) product, vector (cross) product, scalar triple product, vector triple product, direction	20%

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	cosines and direction ratios, Scalar and vector fields, Gradient Divergence and Curl, Directional Derivatives. **Applications**: Application of vector in biological systems, Biomechanics, and biomedical image and data analysis. Integration flow.	

Course Outcomes

COs	Course Outcome Statement	KL
CO1	Apply differential calculus to solve biotechnology-related problems.	K3
CO2	Use integral calculus techniques in biological and biochemical applications.	K3
CO3	Solve differential equations arising in biotechnology systems.	K3
CO4	Apply matrix methods to analyze biological and engineering models.	K3
CO5	Use vector calculus to model and analyze biological systems and transport phenomena.	K3

CO-PO Articulation Matrix

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	2	2	–	–	–	–	–	–	1
CO2	3	3	2	2	2	–	–	–	–	–	–	1
CO3	3	3	2	3	2	–	–	–	–	–	–	2
CO4	3	3	2	2	3	–	–	–	–	–	–	2
CO5	3	3	2	3	3	1	1	–	–	–	–	2

Evaluation Pattern

Sr. No.	Details of the Evaluation	Weightage
1.	Internal Assessment (CT + Assignment + Quiz + Attendance)	30%
2.	External Assessment	70%

Text Books:

1. E. Kreyszig, Advance Engineering Mathematics, John Wiley & Sons, 2005.

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2. B.V. Ramana, Higher Engineering Mathematics, Tata McGraw-Hill Publishing Company Ltd., 2008.
3. B.S. Grewal, Higher Engineering Mathematics, Khanna Publisher, 2005.
4. R.K. Jain & S.R.K. Iyenger, Advance Engineering Mathematics, Narosa Publishing-House, 2002
5. Mathematics - Textbook for Class XI, NCERT Publication
6. Mathematics Part I - Textbook for Class XII, NCERT Publication.
7. Mathematics Part II - Textbook for Class XII, NCERT Publication.

Reference Books:

1. Peter V.O' Neil, Advance Engineering Mathematics, Thomson (Cengage) Learning,2007.
2. Maurice D. Weir, Joel Hass, Frank R. Giordano, Thomas, Calculus, Eleventh Edition, Pearson.
3. G.B Thomas, RL Finney, Calculus and Analytical Geometry, Ninth Edition Pearson, 2002.
4. James Ward Brown and Ruel V Churchill, Fourier Series and Boundary Value Problems,8th Edition-McGraw-Hill
5. D. Poole, Linear Algebra: A Modern Introduction, 2nd Edition, Brooks/ Cole, 2005.
6. Veerarajan T., Engineering Mathematics for first year, McGraw-Hill, New Delhi, 2008.
7. Charles E Roberts Jr, Ordinary Diffrential Equations, Application, Model and Computing, CRC Press T&F Group.
8. Ray Wylie C and Louis C Barret, Advanced Engineering Mathematics, 6th Edition, McGraw-Hill.
9. James Ward Brown and Ruel V Churchill, Complex Variable and Applications, 8th Edition, McGraw-Hill.
10. P. Sivaramakrishna Das and C. Vijaya kumari, Engineering Mathematics, 1st Edition, Pearson India Education Services Pvt. Ltd.
11. Advanced Engineering Mathematics By Chandrika Prasad, Reena Garg Khanna Publishing House, Delhi.
12. Laplace Transforms by Schaum's series, 2005 Edition, Spiegel Publication.
13. Dan Hamilton, Calculus 1 - Differentiation and Integration, Hamilton Education Guides 2018.

Back to Course Structure & Evaluation Scheme

BT Stream

Course Code	**AAS203F**
Title of the Course	Calculus and Numerical Methods
Total Credits of the Course	04
NCrF Hours	60

Main Focus of the Course Outcomes

Employability	**Skill Development**	**Entrepreneurship**
✓	✓	✓

Course Objectives

1. Understand advanced mathematical techniques such as series solutions, integral transforms, complex analysis, and applied linear algebra for solving engineering problems.
2. Apply mathematical methods to model, analyze, and solve problems encountered in biotechnology, biochemical engineering, and life sciences.
3. Develop analytical and computational skills for interpreting biological systems, signal processing, and experimental data.
4. Utilize mathematical tools for research, simulation, optimization, and multidisciplinary applications in biotechnology.

Course Content

Unit	**Description**	**Weightage**
1.	**Series Solution & Special Functions:** Ordinary and singular point, Series solution about an ordinary point, Solution of equation by Frobenius method to solve second order linear differential equation, formation of Indicial equation, Determination of indicial roots, cases of indicial roots: Distinct roots, equal roots, roots differ by an integer and not differ by an integer. Applications: Modeling enzyme kinetics using differential equations, Population growth and microbial growth models	20%
2.	**Laplace Transform:** Definition of Laplace Transform, Existence Conditions, Linearity Property, Laplace Transform of some standard functions, I and II Shifting theorems (without proof), Change of Scale Property, Laplace transform of derivatives and integrals, Periodic Functions, Unit Step Function, Inverse Laplace Transform,	20%

BT Stream

	Partial Fraction Method, Convolution theorem (without proof), Application of Laplace transform to solve ordinary differential equation. **Applications:** Dynamic analysis of bioreactors, Pharmacokinetic and pharmacodynamic modeling.	
3.	**Fourier Series & Fourier Transform:** Periodic function, even and odd function, Dirichlet's Condition for the Existence of a Fourier Series, Fourier series expansion of a function in the interval c to c+2l, Change of Interval, Half Range Series: Half range sine and cosine series in the interval -l to l, Fourier Integral theorem, Fourier transform, Inverse Fourier Transform, Finite Fourier Sin and Cosine transform. **Applications:** ECG and EEG signal analysis, DNA and protein sequence signal processing.	20%
4.	**Numerical Methods I:** Zeroes of transcendental and polynomial equations using Bisection method, Regula-falsi method and Newton-Raphson method, Rate of convergence of above methods. Interpolation: Finite differences, Newton's forward and backward interpolation, Lagrange's and Newton's divided difference formula for unequal intervals. **Applications:** Enzyme Kinetics, Drug Concentration Estimation, Microbial Growth Analysis, Bioprocess Data Analysis	20%
5.	**Numerical Methods II:** Solution of system of linear equations, Matrix Decomposition method, Jacobi method, Gauss- Seidel method. Numerical differentiation, Numerical integration, Trapezoidal rule, Simpson's one third and three-eight rules, Solution of ordinary differential equations by Picard's and fourth-order Runge-Kutta methods. **Applications:** Pharmacokinetic Modeling, Population Growth Models, Nutrient Transport, Numerical Solution of Biological ODEs.	20%

BT Stream

Course Outcomes

COs	Course Outcome Statement	KL
CO1	Apply series solutions and Frobenius method to solve differential equations arising in biotechnology and biological systems.	K3
CO2	Use Laplace Transform techniques to solve differential equations encountered in biochemical processes and dynamic biological models.	K3

CO3	Apply Fourier Series and Fourier Transform techniques for analyzing biological signals and biomedical data.	K3
CO4	Utilize concepts of complex analysis to solve engineering and biological problems involving fluid flow, transport phenomena, and potential functions.	K3
CO5	Apply eigenvalue methods, matrix decompositions, and linear transformations to analyze biological datasets, bioinformatics problems, and computational models.	K3

CO-PO Articulation Matrix

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	2	2	–	–	–	–	–	–	1
CO2	3	3	2	2	2	–	–	–	–	–	–	1
CO3	3	3	2	3	3	–	–	–	–	–	–	2
CO4	3	3	2	3	2	1	1	–	–	–	–	2
CO5	3	3	2	2	3	–	–	–	–	–	–	2

Evaluation Pattern

Sr. No.	Details of the Evaluation	Weightage
1.	Internal Assessment (CT + Assignment + Quiz + Attendance)	30%
2.	External Assessment	70%

Text Books:

1. E. Kreyszig, Advance Engineering Mathematics, John Wiley & Sons, 2005.
2. B.V. Ramana, Higher Engineering Mathematics, Tata McGraw-Hill Publishing Company Ltd., 2008.
3. B.S. Grewal, Higher Engineering Mathematics, Khanna Publisher, 2005.
4. R.K. Jain & S.R.K. Iyenger, Advance Engineering Mathematics, Narosa Publishing-House, 2002

Reference Books:

1. Peter V.O' Neil, Advance Engineering Mathematics, Thomson (Cengage) Learning,2007.

BT Stream

2. Maurice D. Weir, Joel Hass, Frank R. Giordano, Thomas, Calculus, Eleventh Edition, Pearson.
3. G.B Thomas, RL Finney, Calculus and Analytical Geometry, Ninth Edition Pearson, 2002.
4. James Ward Brown and Ruel V Churchill, Fourier Series and Boundary Value Problems,8th Edition-McGraw-Hill
5. D. Poole, Linear Algebra: A Modern Introduction, 2nd Edition, Brooks/ Cole, 2005.
6. Veerarajan T., Engineering Mathematics for first year, McGraw-Hill, New Delhi, 2008.
7. Charles E Roberts Jr, Ordinary Diffrential Equations, Application, Model and Computing, CRC Press T&F Group.
8. Ray Wylie C and Louis C Barret, Advanced Engineering Mathematics, 6th Edition, McGraw-Hill.
9. James Ward Brown and Ruel V Churchill, Complex Variable and Applications, 8th Edition, McGraw-Hill.
10. P. Sivaramakrishna Das and C. Vijaya kumari, Engineering Mathematics, 1st Edition, Pearson India Education Services Pvt. Ltd.
11. Advanced Engineering Mathematics By Chandrika Prasad, Reena Garg Khanna Publishing House, Delhi.
12. Laplace Transforms by Schaum's series, 2005 Edition, Spiegel Publication.

Dan Hamilton, Calculus 1 - Differentiation and Integration, Hamilton Education Guides 2018.

Back to Course Structure & Evaluation Scheme

BT Stream

Course Code	**AEE101C/ AEE201C**
Title of the Course	Fundamentals of Electrical Engineering
Course Category	Engineering Science Course (ESC)
Total Credits	03
NCrF Hours	45

Main Focus of the Course Outcomes

Employability	**Skill Development**	**Entrepreneurship**
✓	✓	✓

Course Objectives:

1. Understand basic electrical quantities and apply Ohm's law and Kirchhoff's laws to simple circuits
2. Understand AC power concepts relevant to biomedical equipment and agricultural machinery
3. Understand the concept of electrical measurements and measuring devices.
4. Learn how transformers and motors work and their applications.
5. Understand electrical safety, earthing, sensors and transducers and relate electrical engineering to emerging/industrial applications.

Course Content

Unit	**Description**	**Weightage**
1.	**Basic Electrical Quantities and DC Circuits:** Electric charge, current, voltage, resistance, power, and energy; Ohm's law; series and parallel resistors; voltage divider; Kirchhoff's Voltage Law (KVL) and Kirchhoff's Current Law (KCL); Simple mesh and nodal analysis;	20%
2.	**AC Circuits and Power:** AC voltage and current — peak, RMS, average values; phasor representation; R, L, and C circuits and their series and parallel connections; admittance and impedance; active power (P), reactive power (Q), apparent power (S); power factor and its significance; single-phase power measurement using wattmeter.	20%

BT Stream

3.	**Electrical Measurements:** Basic working principle of measuring instruments and their types, Definitions of accuracy, precision, resolution, and sensitivity; Types of errors: Gross, systematic, and random errors; DC and AC measurement, Construction and principle of PMMC (Permanent Magnet Moving Coil), Moving Iron (MI), dynamometer, and induction-type instruments	20%
4.	**Transformers and Electrical Machines:** Working principle of single-phase transformer; EMF equation; turns ratio; efficiency and losses; step-up and step-down transformers; DC motor — types, principle, torque, speed control (basic); three-phase induction motor — rotating magnetic field, slip, torque-speed characteristics	20%
5.	**Unit V – Electrical Safety and Emerging Applications:** Electrical hazards — shock and fire; earthing and grounding; working principle of fuses, sensors and transducers; electrical safety in laboratory; **Applications:** Working of fluorescent lamps; motor used in water pumps; key components of energy bills; sensors and transducers in bio-medical and agriculture; three-phase irrigation pump motors.	20%

Teaching-Learning Methodology

Topics are taught using interactive lectures, real-device demonstrations, and examples directly from farming and biotechnology laboratories. Simple numericals on motor power, transformer sizing, and energy billing are the focus of tutorials. Short videos on solar pumps, electric tractors, and smart greenhouse systems are used to connect theory to practice.

Course Outcomes (COs)

COs	Course Outcome Statement	KL
CO1	Apply Ohm's law and Kirchhoff's laws to simple DC circuits in agricultural and biomedical contexts	K3
CO2	Solve single-phase AC circuits and explain power requirements of agri and biomedical equipment	K3
CO3	Understand the concept of electrical measurements and measuring devices.	K3

BT Stream

CO4	Explain fundamental concepts of transformer and machines	K2
CO5	Understand electrical safety, earthing, sensors and transducers and relate electrical engineering to emerging/industrial applications	K2

CO–PO Articulation Matrix

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	1	2	2	–	–	–	–	–	–	1
CO2	3	3	2	2	2	–	1	–	–	–	–	1
CO3	3	3	2	2	2	–	1	–	–	–	–	1
CO4	3	2	1	1	2	–	–	–	–	–	–	1
CO5	3	2	1	1	1	2	2	2	–	–	–	1

Evaluation Pattern

Sr. No.	Details of Evaluation	Weightage
1.	Internal Assessment – Class Tests (CT) + Teacher's Assessment (Assignment / Quiz / Attendance)	30%
2.	End Semester Theory Examination	70%

Suggested Textbooks / Reference Books

1. Basic Electrical Engineering – D.C. Kulshreshtha (McGraw Hill)
2. Fundamentals of Electrical Engineering, 2ed, B. Dwivedi, A. Tripathi, (Wiley & Sons)
3. Principles of Electrical Engineering and Electronics – V.K. Mehta (S. Chand)
4. Electrical Technology Vol. I & II – B.L. Theraja and A.K. Theraja (S. Chand)

Online Resources

1. NPTEL – Basic Electrical Engineering, IIT Bombay / IIT Kharagpur
2. ICAR e-Course – Farm Power, Machinery, and Rural Electrification (free access)
3. Bureau of Energy Efficiency (BEE India) – Agricultural Pumping Systems

Additional References

Relevant articles, videos, and resources on latest developments in:

1. Solar-Powered Irrigation: System Design and Economics

BT Stream

2. Electric Tractors and Autonomous Farm Vehicles
3. Uninterrupted Power Supply (UPS) for Biomedical Equipment

Back to Course Structure & Evaluation Scheme

BT Stream

Course Code	**AEC101C / AEC201C**
Title of the Course	Fundamentals of Electronics Engineering
Course Category	Engineering Science Course (ESC)
Total Credits	03
NCrF Hours	45

Main Focus of the Course Outcomes

Employability	**Skill Development**	**Entrepreneurship**
✓	✓	✓

Course Objectives:

1. Understand diode and transistor basics and their role in biomedical and agri electronic circuits
2. Understand how biosensors, environmental sensors, and agri sensors generate electrical signals
3. Use op-amp circuits to amplify and condition small sensor signals
4. Understand digital basics and microcontroller use in biomedical instruments and agri-automation
5. Relate electronics to real applications — ECG machine, pH meter, glucometer, soil moisture IoT node, solar charge controller

Course Content

Unit	**Description**	**Weightage**
1.	**Diodes, Power Supply, and LED Circuits:** p-n junction diode — forward and reverse bias; diode as rectifier; half-wave and full-wave bridge rectifier; filter capacitor; Zener diode — voltage regulator; LED — working and current-limiting resistor; photodiode and phototransistor (what they do); applications: DC regulated supply for biomedical instruments (pH meter, glucometer), LED indicator in diagnostic equipment, photodetector in pulse oximeter (SpO2), light sensor in greenhouse monitoring	20%
2.	**Transistors and Simple Switching Circuits:** Construction & VI characteristics of BJT, NPN BJT	20%

BT Stream

	— basic concept; transistor in saturation (switch ON) and cut-off (switch OFF); transistor driving a relay or solenoid; Introduction to FETs, MOSFET as a switch (simple idea); optocoupler — isolation concept; applications: relay-driven solenoid valve in drip irrigation, transistor switching buzzer/alarm in biomedical device, MOSFET in solar charge controller switch, optocoupler in ECG isolation circuit (patient safety concept)	
3.	**Sensors, Biosensors, and Signal Conditioning:** Sensors for BT/AG — temperature (thermistor, thermocouple, DS18B20), humidity (capacitive), pH electrode, dissolved oxygen sensor, soil moisture sensor (resistive and capacitive), light sensor (LDR, photodiode); Wheatstone bridge for resistive sensor measurement; op-amp as non-inverting amplifier — why small sensor signals must be amplified; op-amp as comparator — threshold alarm circuit; biopotential signals — ECG, EMG concept (mV level signals, need amplification and filtering); applications: temperature and pH monitoring in fermenter, soil NPK sensor, ECG front-end amplifier concept	20%
4.	**Digital Basics and Microcontroller in BT/AG:** Digital signals vs analog signals; basic logic gates — AND, OR, NOT (truth table, simple application); ADC concept — sensor analog output → digital value read by microcontroller; microcontroller overview (Arduino concept) — analog input pin reads sensor, digital output controls actuator; display — LCD and 7-segment; data logging concept — storing sensor readings for analysis; applications: Arduino reading soil moisture sensor and turning on irrigation pump, digital thermometer, pH display in fermentation monitoring, pulse oximeter display	20%
5.	**Biomedical and Agri Electronics — Emerging Applications:** Biomedical instruments overview — ECG machine block diagram (electrodes → amplifier → ADC → display), pulse oximeter (LED + photodiode + processing), glucometer (biosensor + ADC + display), portable ultrasound (concept); agri-electronics — solar charge controller electronics (block diagram), IoT soil sensor node (sensor +	20%

BT Stream

	microcontroller + Wi-Fi module + cloud), drone electronics overview (ESC, BLDC, GPS module concept); wearable health monitors; precision agriculture electronics — automated irrigation controller, smart greenhouse system	

Teaching-Learning Methodology

Topics are taught using interactive lectures, real instrument demonstrations, and videos. Every concept is linked to a biomedical instrument or agricultural device the student will encounter in their career. Simple calculations are preferred over derivations. Falstad/Tinkercad simulations are used to visualise circuit behaviour.

Course Outcomes (COs)

COs	Course Outcome Statement	KL
CO1	Explain diode and rectifier circuit operation and identify their role in biomedical and agri power supplies	K2
CO2	Explain transistor switching and apply it to relay-driven actuator circuits in BT/AG systems	K3
CO3	Select appropriate sensors and apply op-amp signal conditioning to BT/AG measurement problems	K3
CO4	Interpret digital logic and explain how a microcontroller reads sensors and controls actuators	K2
CO5	Describe the electronics of key biomedical instruments and agri-IoT systems at block diagram level	K2

CO–PO Articulation Matrix

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	1	1	2	–	–	–	–	–	–	1
CO2	3	2	2	1	2	–	–	–	–	–	–	1
CO3	3	3	2	2	3	–	–	–	–	–	–	1
CO4	3	2	2	2	3	–	–	–	–	–	–	1
CO5	3	2	2	2	3	1	2	–	–	–	–	1

Evaluation Pattern

Sr. No.	Details of Evaluation	Weightage
1.	Internal Assessment – Class Tests (CT) + Teacher's Assessment (Assignment / Quiz / Attendance)	30%

BT Stream

2.	End Semester Theory Examination	70%

Suggested Textbooks / Reference Books

1. Electronics: Fundamentals and Applications – D. Chattopadhyay and P.C. Rakshit
2. Biomedical Instrumentation – R.S. Khandpur (TMH)
3. Introduction to Biomedical Equipment Technology – J.J. Carr and J.M. Brown

Online Resources

1. NPTEL – Basic Electronics, IIT Kharagpur
2. NPTEL – Sensors and Actuators (relevant modules)
3. Tinkercad Circuits – tinkercad.com (free Arduino + sensor simulation)

Additional References

Relevant articles, datasheets, and resources on:

1. Biosensors: Principles and Applications in Healthcare and Agriculture
2. Precision Agriculture: IoT Sensor Nodes and Automated Irrigation
3. Wearable Health Electronics: ECG, SpO2, and Fitness Trackers

Back to Course Structure & Evaluation Scheme

BT Stream

Course Code	**ACS101**
Title	Programming Languages
Course Category	Engineering Science Course (ESC)
Credits	03
NCrF Hours	45
Tools	GCC / Code::Blocks / VS Code + Google Colab / Jupyter

Main Focus of the Course Outcomes

Employability	**Skill Development**	**Entrepreneurship**
✓	✓	✓

Course Objectives:

1. Write, compile, and run programs in C and Python
2. Use variables, operators, and control flow to solve simple problems
3. Write and use functions; understand arrays and basic string operations
4. Get familiar with Python built-in data types
5. Apply Python to a small task in an emerging domain — data, automation, or IoT

Course Content

Unit	**Language**	**Topics**	**Weightage**
1.	C	**Programming Basics (C):** 1. Structure of a C program; compilation and execution 2. Data types, variables, constants, operators (arithmetic, relational, logical) 3. Input/output — scanf/printf in C 4. Type conversion; simple expressions *Real connect: C powers embedded systems, microcontrollers, and OS kernels.*	20%
2.	C	**Decision Making and Control Flow (C):** 1. Control Statements 2. Loops — for, while, do-while; break and continue 3. Functions — definition, call, return; call by value	20%

BT Stream

		4. Recursion — factorial and Fibonacci (simple examples only) *Real connect: Control flow is the brain of every program — from traffic signal logic to drone autopilot decision-making.*	
3.	C	**Advanced Programming Concepts (C):** 1. 1D array — declaration, input, output, traversal 2. Simple operations — find max, min, sum, average 3. Strings in C — gets/puts, strlen, strcpy, strcmp 4. Pointer — concept of address and pointer variable (concept only, not deep) 5. Structures, 6. File Handling *Real connect: Arrays store sensor readings, image pixels, and stock prices. Pointers are the foundation of memory management in embedded systems.*	20%
4.	Python	**Python Introduction:** 1. Why Python — where it is used (AI, Data Science, Web, Automation) 2. Python basics — variables, data types, print, input, operators 3. Python control flow — if/else, for loop, while loop 4. Python functions — def, return, default arguments *Real connect: Python is #1 in AI/ML, Data Science, and automation — same syntax, more power, less boilerplate.*	20%
5.	Python	**Python Data Structures and Emerging Applications** 1. Python list, tuple, dictionary, set — basic operations only 2. File handling — read and write a text file; read a CSV (2–3 lines) 3. Introduction to NumPy — array, mean, simple arithmetic (5 lines) 4. Introduction to Matplotlib — plot a simple line graph (5 lines) 5. One mini-demo — IoT sensor simulation	20%

BT Stream

		Real connect: Python lists power NumPy which powers TensorFlow. Matplotlib plots are used in every engineering report and data science notebook. This unit opens the door to AI, IoT, and Data Science using the same Python basics learnt in Unit IV.	

Tutorial Hour — Week-by-Week Practice Problems

One tutorial hour per week. Students write and run programs. Problems below are the minimum set — teacher may add variants.

Week	Language	Tutorial Problems
1	C	1. Print your name and branch. 2. Add two numbers entered by user. 3. Find area of a rectangle.
2	C	1. Swap two numbers. 2. Check if a number is even or odd. 3. Simple calculator (+, –, ×, ÷) using if-else.
3	C	1. Find the largest of three numbers. 2. Grade calculator (marks → A/B/C/D/F). 3. Day name using switch-case.
4	C	1. Print multiplication table of N. 2. Sum of first N natural numbers using loop. 3. Count digits in a number.
5	C	1. Print prime numbers from 1 to 50. 2. Fibonacci series (first 10 terms). 3. Reverse a number.
6	C	1. Factorial using function. 2. Check palindrome number using function. 3. Find GCD of two numbers.
7	C	1. Recursive factorial. 2. Recursive Fibonacci. 3. Sum of digits using recursion.
8	C	1. Input array of N numbers; find max and min. 2. Compute average of array. 3. Linear search in array.
9	C	1. Count vowels in a string. 2. Reverse a string. 3. Check if string is palindrome.
10	C	1. Write a C program to store 5 numbers in an array and print them in reverse order. 2. Write a C program to perform basic string operations such as finding string length, copying and concatenation.
11	Python	1. Hello World in Python. 2. BMI calculator. 3. Temperature converter (°C ↔ °F). Compare C vs Python syntax.

BT Stream

12	Python	1. Grade calculator in Python. 2. Fibonacci using Python loop. 3. Simple calculator using functions in Python.
13	Python	1. List: marks of 5 students — max, min, average. 2. Dictionary: store name and marks of 3 students. 3. Count word frequency in a sentence.
14	Python	1. Write student records to a text file. 2. Read and display the file. 3. Read a small CSV and print rows.
15	Python	1. NumPy: create array, compute mean. 2. Matplotlib: plot a line graph of marks. 3. Mini-demo: simulate 5 IoT temperature readings and plot.

Teaching-Learning Methodology

Lectures are short and concept-focused — every idea is shown with a 3–5 line code example. Tutorial hour is compulsory hands-on coding. Students use GCC/Code::Blocks for C and Google Colab for Python (no installation needed for Python). Live coding and debugging by the teacher are part of every session. Errors are learning opportunities — never skip them.

Course Outcomes (COs)

COs	Course Outcome Statement	KL
CO1	Write, compile, and run C programs using variables, operators, and I/O	K3
CO2	Use control flow and functions in C to solve simple computational problems	K3
CO3	Work with arrays and strings in C and understand pointer concept	K2
CO4	Implement basic operations using Python built-in data structures.	K3
CO5	Apply Python to read files, plot data, and perform a simple emerging domain task	K3

CO–PO Articulation Matrix

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	1	1	3	–	–	–	–	–	–	1
CO2	3	3	2	2	3	–	–	–	–	–	–	1
CO3	3	2	1	2	3	–	–	–	–	–	–	1
CO4	3	3	2	2	3	–	–	–	–	–	–	1

BT Stream

CO5	3	3	2	2	3	1	1	–	–	–	–	2

Evaluation Pattern

Sr.	Component	Marks	Weightage
1.	Class Tests (CT) — short questions + write a program (C/Python)	15	15%
2.	Teacher's Assessment (TA) — Tutorial performance + Assignments + Quiz	15	15%
3.	End Semester Examination — theory + program writing (both languages)	70	70%

Suggested Textbooks

1. Problem Solving and Programming in C, Sridhar & Hariharan, Pearson
2. Basic Computer Engineering, 2ed, Sanjay Silakari, Rajesh K. Shukla, Wiley & Sons
3. Python Programming, Sridhar, Indumathi & Hariharan, Pearson
4. Let Us C – Yashavant Kanetkar (BPB) — for C basics
5. Python Crash Course – Eric Matthews (No Starch Press) — for Python units

Online Resources

1. OnlineGDB – onlinegdb.com (free browser-based C and Python compiler)
2. Google Colab – colab.research.google.com (free Python + NumPy + Matplotlib, no install)
3. HackerRank – hackerrank.com (C, Python practice, free, auto-graded)
4. NPTEL – Programming in C, IIT Bombay (free video lectures)

Back to Course Structure & Evaluation Scheme

BT Stream

Course Code	**ACS201**
Title of the Course	Essentials of Data Structure
Pre-requisite	ACS101 – Programming Language (C/C++ + Python)
Course Category	Engineering Science Course (ESC)
Credits	02
NCrF Hours	30

Main Focus of the Course Outcomes

Employability	**Skill Development**	**Entrepreneurship**
✓	✓	✓

Course Objectives:

1. Understand data structures and why choosing the right one affects program performance
2. Implement linked list and perform basic insert, delete, and traverse operations
3. Implement stack and queue and apply them to simple real-world problems
4. Apply searching and sorting algorithms and compare their efficiency
5. Understand trees and graphs at a conceptual level with real-world connections

Course Content

Unit	**Language**	**Topics**	**Weightage**
1.	C / Python	**Introduction and Arrays** 1. Data structure — definition, need, classification (linear / non-linear, static / dynamic) 2. Algorithm efficiency — Big O notation: O(1), O(n), O(log n), O(n^2) — concept only 3. Array — insert, delete, search, traverse in C/Python 4. 2D array — matrix addition and transpose *Real connect:NumPy array (AI/ML), image as 2D matrix, sensor data log as array, spreadsheet rows.*	20%

BT Stream

2.	C/Python	**Linked List** 1. Node — data + pointer; why linked list over array 2. Singly linked list — insert (head, tail, position), delete, traverse 3. Doubly linked list — concept and structure only 4. Circular linked list — concept only *Real connect: Browser back/forward = doubly linked list. Music playlist = circular. Python deque = doubly linked list (used in BFS, LRU cache).*	20%
3.	C/Python	**Stack and Queue** 1. Stack — LIFO; push, pop, peek; implementation using array in C/Python 2. Application: balanced parentheses check; function call stack (concept) 3. Queue — FIFO; enqueue, dequeue; implementation using array in C/ Python 4. Application: print spooler; task scheduling queue, Polish Notation *Real connect: Ctrl+Z = Stack. WhatsApp delivery = Queue. Python error traceback IS a call stack. CPU scheduling = queue.*	20%
4.	C/Python	**Searching and Sorting** 1. Linear search, binary search, comparison 2. Bubble sort and insertion sort, Quick Sort and Selection Sort — trace + C/ Python 3. Merge sort — concept and trace only 4. Python built-in sort() and sorted() *Real connect: Binary search in database index. Python sort() uses Timsort. E-commerce ranking, leaderboard, ML data preprocessing = sorting.*	20%
5.	Trace + Python demo	**Trees and Graphs (Concept Level)** 1. Binary tree — terminology (root, leaf, height, level); properties 2. Binary Search Tree — insert, search; inorder/preorder traversal trace 3. Graph — vertex, edge, directed/undirected; adjacency list representation, insertion	20%

BT Stream

		4. BFS and DFS — trace on a small graph (5–6 nodes) *Real connect: File system = tree. BST in database indexing. Google Maps = weighted graph. Social network = graph. Decision Tree in ML = binary tree.*	

Suggested Assignment / Practice Problems (Unit-wise) but not limited to

No Tutorial hour in ACS201. The following problems are assigned as class assignments and self-practice. 2–3 per unit recommended for internal assessment.

Unit	Trace on Paper	Code (C/Python)
I Arrays	1. Trace insertion at index 2 in [10,20,30,40] 2. Trace deletion of element 30 from [10,20,30,40] 3. What is O(log n)? Give one example	1. Input N numbers; find max, min, average 2. Count even and odd in array 3. Remove duplicates from array
II Linked List	1. Draw linked list after inserting 10, 20, 30 at head 2. Trace deletion of node 20 from [10→20→30→NULL] 3. Difference between singly and doubly linked list	1. Create linked list; insert at head; print 2. Insert node at tail 3. Delete node by value
III Stack & Queue	1. Trace push(5), push(10), pop(), push(15) — what is top? 2. Trace enqueue(A,B), dequeue(), enqueue(C) — what is front? 3. Check '({[]})' balanced — show stack trace	1. Stack using array — push and pop 5 elements 2. Queue using array — enqueue and dequeue 3. Balanced parentheses checker
IV Search& Sort	1. Binary search for 35 in [10,20,30,35,40,50] — show steps 2. Bubble sort trace on [64,34,25,12] 3. Insertion sort trace on [5,3,8,1]	1. Linear search and binary search — compare 2. Bubble sort — count swaps 3. Python: sort a list ascending and descending
V Tree & Graph	1. Draw BST after inserting 50,30,70,20,40 2. Write inorder and preorder of above BST 3.	1. BST insert and inorder in python 2. Graph as adjacency list in Python dict 3. BFS using queue in Python (10–15 lines)

BT Stream

	BFS from A on: A-B, A-C, B-D, C-E	

Teaching-Learning Methodology

Draw first, code later — every data structure is shown visually on the board before any code is written. Trace problems on paper before coding. VisuAlgo animations are used in class for sorting and tree/graph traversal. Python is used for quick 5-line demos where it adds clarity. Assignments include both trace and coding tasks for every unit.

Course Outcomes (COs)

COs	Course Outcome Statement	KL
CO1	Explain data structure types and Big O; implement array insert, delete, and search in C/Python	K3
CO2	Implement singly linked list operations and relate to real-world applications	K3
CO3	Implement stack and queue using arrays and apply them to solve simple problems	K3
CO4	Apply linear and binary search; trace and implement bubble and insertion sort	K3
CO5	Trace BST insert and traversal; trace BFS/DFS on a small graph	K2

Evaluation Pattern

Sr. No.	Component	Weightage
1.	Internal Assessment – Class Tests (CT) + Assignments / Quiz (TA)	30%
2.	End Semester Examination — theory + trace problems + short code writing	70%

Suggested Textbooks / Reference Books

1. Data Structures Using C – ReemaThareja (Oxford University Press)
2. Data Structures and Algorithms in Python – Goodrich, Tamassia, Goldwasser(Wiley)

BT Stream

3. Problem Solving with Algorithms and Data Structures using Python – Brad Miller (free: runestone.academy)

Online Resources

1. VisuAlgo – visualgo.net (free animations of every DS and algorithm — use in every class)
2. HackerRank – Data Structures track, hackerrank.com (C++/Python, free, auto-graded)
3. Runestone Academy – runestone.academy (free interactive Python DSA book)
4. LeetCode Easy – leetcode.com (start after this course; industry-standard placement prep)

Back to Course Structure & Evaluation Scheme

BT Stream

Course Code	**AAS104/ AAS204**
Title of the Course	Introduction to Indian Knowledge System
Course Category	Engineering Science Course (ESC)
Total Credits	02
NCrF Hours	30

Main Focus of the Course Outcome

Employability	**Skill Development**	**Entrepreneurship**
✓	✓	

Course Objectives

1. To introduce the meaning, scope, classification and significance of the Indian Knowledge System (IKS) and its corpus.
2. To develop an understanding of India's contributions to philosophy, mathematics, astronomy and the sciences.
3. To familiarise students with indigenous engineering, technology, metallurgy and architecture.
4. To create awareness of Indian traditions of health, wellness, environment, governance and ethics.
5. To expose students to the contemporary relevance of IKS — digital preservation, AI-enabled study and sustainable applications.

Course Content

Unit	**Description**	**Weightage**
1	**Foundations of the Indian Knowledge System** Definition, scope, characteristics and significance of IKS; the IKS vision of NEP 2020; the fourteen (Chaturdasha) Vidyasthanas — four Vedas, four Upavedas and six Vedangas; Schools of Indian Philosophy: Orthodox and Heterodox Schools; Oral and Written Knowledge Systems, Knowledge validation (pramanas). Case study: relevance of IKS for a present-day engineer.	20
2	**Indian Mathematics, Astronomy and Scientific Thinking** Number Systems: Decimal place-value system, zero and infinity; Bhutasankhya and Katapayadi numeration; Pingala's Chandahshastra and binary	20

BT Stream

	representation; Sulbasutras; contributions of Aryabhata, Brahmagupta, Bhaskara and the Kerala School; Indian astronomy and the Panchanga. Numerical: encoding numbers using the Katapayadi scheme/Observing Panchang and Reflecting on the Patterns	
3	**Engineering, Technology and Material Sciences** Metallurgy — Wootz steel, zinc distillation and the Delhi Iron Pillar; town planning and architecture (Harappan, Vastu and Shilpa Shastra); water management — stepwells, tanks and irrigation, textile engineering Case study: materials-science analysis of the Iron Pillar/ Architecture of Stupas	20
4	**Health, Wellness, Environment and Governance** Ayurveda (Charaka and Sushruta), Tridosha; Yoga (Patanjali's Ashtanga) and wellness; environment and sustainability in Indian thought; governance and management — Kautilya's Arthashastra; ethics from the Bhagavad Gita. Case study: Yoga for well-being / Arthashastra management principles/ Learnings from Geeta	20
5	**Contemporary Relevance, AI and Applications of IKS** Digital preservation (CSIR-TKDL, IKS Division) and IPR; Sanskrit and computational linguistics — Panini's Ashtadhyayi, NLP and AI for ancient texts; IKS-inspired sustainable engineering and innovation. Case study: AI/NLP applied to a Sanskrit corpus.	20

BT Stream

Teaching-Learning Methodology

Delivery through interactive lectures, documentaries and IKS Division resources, expert talks, heritage field visits, assignments and a mini-project connecting IKS to a contemporary problem, supported by NPTEL / SWAYAM MOOCs, in line with NEP 2020 and OBE.

Course Outcomes (COs)

COs	Course Outcome Statement	KL
CO1	Learn the nature, scope and classification of the Indian Knowledge System.	K1
CO2	Understand India's contributions to mathematics and astronomy and relate them to modern concepts.	K2

CO3	Understand indigenous engineering, metallurgical and architectural achievements.	K2
CO4	Interpret Indian traditions of health, wellness, governance and ethics.	K2
CO5	Recognize the contemporary relevance of IKS, including AI-enabled study and sustainability.	K2

CO-PO Articulation Matrix

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	1					2	1	2	1	1		3
CO2	2	1	1		1	1		1				2
CO3	2	2	2		1	2	2	1				2
CO4						2	2	3	1	1		3
CO5	1	1	2		2	3	3	2	1	1	1	3

Evaluation Pattern		
Sr. No.	**Details of the Evaluation**	**Weightage**
1	Internal Continuous Assessment — Class Tests, Assignments, Quizzes, Seminars and Attendance (CT + TA)	30 %
2	End-Semester Examination (Theory)	70 %

Suggested References (latest editions)	
1	Mahadevan, Bhat & Nagendra Pavana, Introduction to Indian Knowledge System: Concepts and Applications, PHI Learning.
2	Kapil Kapoor & Avadhesh Kumar Singh (Eds.), Indian Knowledge Systems, Vols. I & II, D. K. Printworld.
3	Pride of India: A Glimpse into India's Scientific Heritage, Samskrita Bharati.
4	B. V. Subbarayappa, Science in India: Past and Present, Nehru Centre.
Online / e-Resources	
1	IKS Division portals — iksindia.org and iks.aicte-india.org.
2	SWAYAM / NPTEL course on Indian Knowledge Systems; CSIR-TKDL; NDLI; Bharatavani.

Back to Course Structure & Evaluation Scheme

Course Code	**AAS105 / AAS205**
Title of the Course	Professional Communication and Technical Writing
Course Category	HSMC
Total Credits	02
NCrF Hours	30

Main Focus of the Course Outcomes

Employability	**Skill Development**	**Entrepreneurship**
✓	✓	

Course Objectives

1. Explain the principles of professional communication, workplace etiquette, and business correspondence required for professional environments.
2. Apply verbal, non-verbal, presentation, and AI-assisted communication strategies to communicate effectively in academic and profession.
3. Analyze workplace situations by using emotional intelligence, critical thinking, and ethical decision-making based on Indian Knowledge Systems.
4. Demonstrate leadership, teamwork, interpersonal communication, and collaborative problem-solving skills in diverse professional contexts.
5. Integrate mindfulness, yogic principles, and professional ethics to promote effective communication, personal well-being, and lifelong professional growth.

Unit	**Description**	**Weightage**
1	**Professional Communication and Workplace Readiness** • Fundamentals of communication: definition, process, importance of communication barriers to communication, 7 C's of communication • Business Correspondence : Emails, Notices, Agenda, Minutes of Meeting, Technical Report and proposal • Résumé, CV and Cover Letter Writing	20

BT Stream

	• Interview Skills, Professional Networking and branding e,g.,LinkedIn , X, ATS Score • Digital Communication Etiquette : online presence and virtual meetings **Assignments/Activities** • Prepare a professional résumé and cover letter. • Draft business emails and meeting minutes. • Participate in a mock interview.	
2	**Grammar and Vocabulary for Technical Writing** • Parts of Speech and Sentence Structure • Common Grammatical Errors, Subject–Verb Agreement, Tenses, and Articles • Technical Vocabulary, Word Formation, and Precision in Language • Tone, Style, and Clarity in Technical English • Editing and Proofreading Techniques for Technical Writing • Grammar and Vocabulary Enhancement through Practical Writing Exercises **Assignments/Activities** • Edit and proofread a technical paragraph to improve grammar, vocabulary, precision, and clarity.	20
3	**Public – Speaking, Presentation and AI-Assisted Communication** • Public speaking technique: Modes, verbal, and non-verbal communication • Understating non-verbal Communication through the *Natyashastra: Angika Abhinaya* (Body Language: Posture, Gestures, Facial Expressions),*Vachika Abhinaya* (Voice Modulation and Speech Delivery) ,*Sattvika Abhinaya* (Expression of Emotions and Authentic Communication) • Presentation Planning and Delivery • Data Visualization and Slide Design • Elevator Pitch • AI Tools for Presentation Design and Speech Practice, prompt writing for effective communication • Ethical Use of AI in Communication **Assignments/Activities** • Deliver an individual presentation. • Prepare an AI-assisted presentation with proper citations. • Record and evaluate an elevator pitch.	20

BT Stream

4	**Emotional Intelligence and Critical Thinking for Workplace Communication** • Emotional Intelligence through the *Bhagavad Gita* :Self-awareness ,Self-regulation ,Empathy, Emotional Balance • Critical Thinking through Indian Values and Logical Traditions: *Nyaya* (Logical Reasoning), *Viveka* (Discrimination and Judgment), • Value-based Decision Making • Ethical Problem Solving • Reflective Thinking in Professional Contexts **Assignments/Activities** • Case study analysis using value-based decision making. • Reflective journal on emotional intelligence. • Group discussion on workplace ethical dilemmas.	20
5	**Leadership, Workplace Well-being, and Mindful Communication** • Leadership definition and types • Leadership Communication: Active Listening, Constructive Feedback, Teamwork, and Collaboration • Mental Health, Stress Management, and Work-Life Balance • Effective Communication through Yogic Practices: **Yama** as Principles of Ethical Communication **Ahimsa** – Respectful and Non-violent Communication, **Satya** – Truthfulness and Transparency, **Asteya** – Integrity and Honesty, **Brahmacharya** – Responsible and Balanced Communication, **Aparigraha** – Freedom from Ego, Bias, and Possessiveness. **Niyama** for Personal and Professional Growth, **Shaucha** – Clarity of Thought and Communication, **Santosha** – Contentment and Positive Workplace Relationships, **Tapas** – Self-discipline and Professional Commitment, **Svadhyaya** – Self-reflection and Continuous Learning ,**Ishvara Pranidhana** – Purpose, Humility, and Service-oriented Leadership • **Storytelling through Panchatantra:** *The Lion and the Rabbit* (Leadership and Strategic Communication), *The Merchant's Iron Balance* (Emotional Intelligence and Strategic Conflict Resolution), *The Two Fish and the Frog* (Leadership,	20

BT Stream

	Vigilance and Decision Making), and *The Merchant and the King's Servant* (Respectful Leadership, Workplace Ethics and Employee Relations).	

Teaching-Learning Methodology	Sessions will be activity-based and interactive, combining short lectures with role-plays, presentations, group discussions, writing tasks and peer review. Audio-visual material, language-lab practice and NPTEL/SWAYAM resources will develop practical communication skills in line with NEP 2020 and OBE.

Course Outcomes (COs)

COs	Course Outcome Statement	KL
CO1	Communicate effectively in professional settings using appropriate verbal, written, and digital communication skills.	K2
CO2	Apply correct grammar and vocabulary to produce clear technical English.	K3
CO3	Design and Deliver effective presentations by applying public speaking techniques and AI-assisted communication tools ethically.	K4
CO4	Apply emotional intelligence, logical reasoning, and value-based decision-making to solve workplace challenges.	K3
CO5	Demonstrate leadership, teamwork, ethical and mindful communication, stress management, and professional well-being for effective and sustainable workplace success.	K6

CO–PO Articulation Matrix

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO11	PO12
CO1	1	–	–	–	1	–	–	–	2	3	2	2	2
CO2	1	1	–	–	1	–	–	–	–	3	2	2	2
CO3	1	2	1	–	2	–	–	2	2	3	2	2	2
CO4	1	3	–	–	–	2	–	3	2	2	2	2	2
CO5	–	2	–	–	–	2	–	3	3	3	3	3	3

Evaluation Pattern		
Sr. No.	**Details of the Evaluation**	**Weightage**
1	Internal Continuous Assessment Class Tests, Assignments, Quizzes, Seminars and Attendance (CT + TA)	30 %

BT Stream

2	End-Semester Examination (Theory)	70 %

Online / e-Resources
Meenakshi Raman & Sangeeta Sharma, Technical Communication: Principles and Practice, Oxford University Press.
Vidushi Sharma , Advanced Soft Skills , Pearson
R. C. Sharma & Krishna Mohan, Business Correspondence and Report Writing, McGraw Hill.
Lesikar & Flatley, Business Communication, McGraw Hill.
NPTEL / SWAYAM courses on Technical / English Communication and Soft Skills.
British Council Learn English and similar e-resources for language practice.
Bhagavad Gita : https://ignca.gov.in/Asi_data/279.pdf
Patanjali. The Yoga Sutras of Patanjali https://archive.org/details/yogasutrasofpata00pata/page/28/mode/2up https://www.ebharatisampat.in/pdfs/ebharati-pdf-1628160725PatanjalaYogaSutra-The-Yoga-Aphorisms-Patanjali-1953.pdf
Natyashastra: https://archive.org/details/natyashastraofbharatmunipart1pradipahindivyakhyababulalshuklashastrichowkambhasanskritseries/page/%E0%A5%A7%E0%A5%AA%E0%A5%AB/mode/2up
Panchatantra : https://ignca.gov.in/Asi_data/2332.pdf https://www.talesofpanchatantra.com/sanskrit-manuscripts https://www.peoplematters.in/blog/organisational-culture/panchatantras-for-leaders-13994
Critical thinking : https://www.alexpcherian.com/2025/08/nyaya-philosophy-art-of-logical-thinking.html https://jiva.org/viveka-proper-discrimination/

Back to Course Structure & Evaluation Scheme

BT Stream

Course Context: AI is no longer optional — it is the defining technology of this decade. Every engineer, regardless of branch, will work alongside AI tools in their career. This course demystifies AI for first-year students from all streams: what it is, how it works at a concept level, where it is already changing their field, and — most practically — how to communicate with AI systems effectively through Prompt Engineering. No prior programming or mathematics required. Concept, awareness, and hands-on practice are the goals.

Course Code	**AAI101/AAI201**
Title of the Course	Introduction to AI and Prompt Engineering
Course Category	Engineering Science Course (ESC)
Credits	02
NCrF Hours	30
Tools Used	ChatGPT / Claude / Gemini / Copilot (free tier) — browser based

Main Focus of the Course Outcomes

Employability	**Skill Development**	**Entrepreneurship**
✓	✓	✓

Course Objectives:

1. Understand what Artificial Intelligence is, its history, and how it is changing every engineering field
2. Understand key AI concepts — Machine Learning, Deep Learning, and Neural Networks — at a conceptual level
3. Explore how AI is applied across domains — healthcare, agriculture, manufacturing, finance, and education
4. Understand Generative AI — how large language models (LLMs) like ChatGPT and Gemini work
5. Learn and practise Prompt Engineering — the skill of communicating effectively with AI tools to get useful results

Course Content

Unit	**Topics**	**Weightage**
1.	**Introduction to Artificial Intelligence** 1. Definition and scope of AI; brief history — from Alan Turing to ChatGPT 1. Types of AI — Narrow AI, General AI, Super AI (concept)	20%

BT Stream

	2. AI vs Machine Learning vs Deep Learning — what is the difference 3. How machines learn — data, patterns, and predictions (concept, no maths) 4. AI today — where it already exists in daily life (Google Maps, Spotify, Netflix, Google Photos, voice assistants) 5. Machine Learning Concepts, Learning Strategies- Supervised, Unsupervised, Reinforcement Learning 6. Data, Data Visualization, Data Pre-processing **Class Activity: Students list 5 AI tools they use every day and discuss how each one 'learns'.** *Branch connect: CE — AI in structural health monitoring; ME — AI in predictive maintenance; EE — smart grid; CSE — AI in compilers; BT — drug discovery; AG — crop yield prediction.*	
2.	**Core AI Concepts** 1. Machine learning: Classification Techniques- K-Nearest Neighbors (KNN), Naïve Bayes, Clustering Technique-K-Means 2. Classification metrics-accuracy, precision, recall, F1 score, MAE. 3. Neural networks — concept of neurons, layers, weights (simple diagram) 4. Deep learning — what makes it 'deep'; why it needs GPUs, CNN 5. Computer vision — how AI sees images (face recognition, medical imaging) 6. Natural Language Processing (NLP) — how AI understands text and speech **Class Activity: Show Google Teachable Machine (teachablemachine.withgoogle.com) — train a simple image classifier in 5 minutes, no code.** *Branch connect: ECE — image sensors + AI; BT — protein structure prediction (AlphaFold); AG — plant disease detection from photos; TX — fabric defect detection.*	20%
3.	**Generative AI and Large Language Models** 1. What is Generative AI — AI that creates text, images, code, audio, video 2. Large Language Models (LLMs) — what they are, how they are trained (concept)	20%

BT Stream

	3. Popular LLMs — ChatGPT (OpenAI), Gemini (Google), Claude (Anthropic), Llama (Meta) 4. Tokens, context window, temperature — simple explanation 5. Multimodal AI — text + image + voice in one model (GPT-4o, Gemini concept) 6. AI image generation — Stable Diffusion, Midjourney, DALL-E (concept + demo) 7. Limitations of LLMs — hallucination, knowledge cutoff, bias 8. Responsible use — academic integrity, plagiarism, AI-generated content disclosure **Class Activity: Students interact with ChatGPT/Gemini/Claude; observe how different inputs give different outputs; identify one hallucination example.** *Every professional today uses at least one Generative AI tool. Understanding how LLMs work makes students better users and future developers.*	
4.	**AI Applications Across Engineering Domains** 1. AI in Civil Engineering — structural health monitoring, smart cities, traffic optimisation 2. AI in Mechanical Engineering — predictive maintenance, design optimisation, robotics 3. AI in Electrical and Electronics — smart grids, fault detection, VLSI design automation 4. AI in CSE — code generation (GitHub Copilot), cybersecurity, natural language interfaces 5. AI in Biotechnology and Healthcare — drug discovery, medical imaging, genomics, wearables 6. AI in Agricultural Engineering — precision farming, crop monitoring, drone-based spraying, soil analysis 7. AI in Chemical and Textile Engineering — process optimisation, quality control, material design 8. Ethical concerns — bias in AI, data privacy, job displacement, responsible AI **Class Activity: One case study per student group — each group presents a 2-minute real AI application from their own engineering branch.** *This unit ensures every student, regardless of branch, sees AI as directly relevant to their future career.*	20%

Bt Stream

5.	**Prompt Engineering** 1. What is a prompt — the instruction you give to an AI model 2. Why prompt quality matters — same AI, very different results with different prompts 3. Principles of good prompting — clarity, context, specificity, role assignment, format instruction 4. Prompting techniques: • Zero-shot prompting — direct question • Few-shot prompting — give examples in the prompt • Chain-of-thought prompting — ask AI to think step by step • Role prompting — 'Act as a civil engineer and…' 5. Prompt for different tasks — summarisation, explanation, code generation, report writing, brainstorming, translation 6. Domain-specific prompting — students write prompts relevant to their own branch (CE, ME, BT, AG, etc.) 7. AI tools for engineers — GitHub Copilot (code), Grammarly (writing), Consensus (research), Wolfram Alpha, Canva AI, Notion AI **Hands-on Practice (every lecture): Students write prompts, compare outputs, refine and improve. Graded prompt assignment: write 5 prompts for an engineering problem from their own branch.** *Prompt Engineering is already a job role (avg. salary $100K+ globally). Every engineer who can communicate well with AI has a significant career advantage.*	20%

Suggested Assignments / Practice Tasks but not limited to

Assignment	Task
Assignment 1 (Unit I)	Research task — no AI tool interaction required. Find 3 real AI applications in your engineering branch. For each: name the product/system, what data it uses, and what it predicts or generates. Submit as a one-page note (250–300 words).
Assignment 2 (Unit II)	Concept-check task. Pick one real ML-based system relevant to your branch (e.g., a recommendation engine, a defect detector, a chatbot). Identify: (a) Is it supervised, unsupervised, or reinforcement learning? (b) What could cause it to overfit or underfit? (c) Sketch a simple diagram

BT Stream

	showing its inputs, a basic neuron/layer concept, and output. Submit as a half-page note (200–250 words) with diagram.
Assignment 3 (Unit III)	Hands-on task — direct AI interaction required. Interact with any free LLM (ChatGPT/Gemini/Claude). Ask it a question related to your branch. Identify one correct and one incorrect (hallucinated) response. Explain why the hallucination occurred (150–200 words), with screenshots of both responses.
Assignment 4 (Unit IV)	Group case study: Each group (3–4 students) picks one AI application in their domain, prepares a 2-minute presentation covering: problem → data → AI solution → impact. Submit slides/notes plus a 1-page group summary handout.
Assignment 5 (Unit V)	Write 5 prompts for an engineering problem from your branch using: (a) Zero-shot (b) Few-shot (c) Chain-of-thought (d) Role prompting (e) Format-specific prompt. Submit the prompt + AI response + your evaluation of the output quality, as a single document (2–3 pages).
Assignment 6 (Unit V)	Use an AI tool to help complete one academic task (summarise a topic, generate a study plan, explain a concept, translate a text). Document the prompt used, output received, and how you verified/corrected it. In a 200–300 word reflection, address: (a) what you changed or verified and why, and (b) one way AI use in this task could raise a bias, privacy, or academic-integrity concern.

BT stream

Teaching-Learning Methodology

Every lecture begins with a real-world AI example — a news article, a product demo, or a live tool interaction. No heavy mathematics or programming — concepts are taught through analogies, diagrams, videos, and demonstrations. Unit V (Prompt Engineering) is entirely hands-on — students bring their laptops/phones and write prompts live in class. Group activities, branch-specific case studies, and AI tool explorations keep every session engaging. Teachers are encouraged to use AI tools themselves in class to demonstrate live prompting and critical evaluation of AI outputs.

Course Outcomes (COs)

COs	Course Outcome Statement	KL
CO1	Explain what AI is, its types, and identify AI applications in everyday life and their own engineering domain	K2
CO2	Describe key AI concepts — ML, Deep Learning, NLP, Computer Vision — and relate them to real applications	K2
CO3	Apply the working principles, capabilities, limitations, and responsible use of Generative AI and Large Language Models.	K3
CO4	Apply the concept of Artificial Intelligence across engineering domains and their societal, ethical, and professional implications.	K3
CO5	Design and refine effective prompts using zero-shot, few-shot, chain-of-thought, and role prompting techniques for domain-specific tasks	K3

CO–PO Articulation Matrix

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	1	–	2	–	–	–	–	–	–	1
CO2	3	2	2	1	3	–	–	–	–	–	–	1
CO3	3	3	2	2	3	1	1	3	–	–	–	2
CO4	3	3	2	2	2	2	2	3	1	1	–	2
CO5	3	3	3	2	3	1	–	2	–	2	–	3

Evaluation Pattern

Sr.	Component	Marks	Weightage
1.	Class Tests (CT) — concept questions + short answer	15	15%
2.	Teacher's Assessment (TA) — Assignments + Group Case Study + Prompt Portfolio	15	15%
3.	End Semester Theory Examination — concept + application + prompt writing questions	70	70%

Suggested Textbooks / Reference Books

1. *The Quick Guide to Prompt Engineering*, I. Khan, John Wiley & Sons Inc.

BT Stream

2. AI: Its Nature and Future – Margaret Boden (Oxford University Press) — accessible, non-technical introduction to AI concepts
3. Human Compatible: Artificial Intelligence and the Problem of Control – Stuart Russell (Penguin) — accessible reading
4. The Age of AI – Henry Kissinger, Eric Schmidt, Daniel Huttenlocher — for AI and society context
5. Prompt Engineering for Generative AI – James Phoenix and Mike Taylor (O'Reilly) — practical guide to LLMs and prompting, supports Units IV & V

Online Resources

1. Google Teachable Machine – teachablemachine.withgoogle.com (train AI with no code, free)
2. Elements of AI – elementsofai.com (free, beginner course by University of Helsinki + MinnaLearn)
3. Microsoft AI for Everyone – learn.microsoft.com (free beginner modules)
4. ChatGPT – chat.openai.com / Gemini – gemini.google.com / Claude – claude.ai (free tier for prompt practice)
5. Prompt Engineering Guide – promptingguide.ai (free, comprehensive reference)
6. Ethics of AI – ethics-of-ai.mooc.fi (free, beginner course by University of Helsinki on bias, privacy, and responsible AI)
7. 3Blue1Brown – Neural Networks series (YouTube, free) — visual, no-maths-heavy explanation of how neural networks work

Back to Course Structure & Evaluation Scheme

BT Stream

Course Code	**AME101C/AME201C**
Title of the Course	Fundamentals of Mechanical Engineering
Course Category	Engineering Science Course (ESC)
Total Credits	02
NCrF Hours	30

Main Focus of the Course Outcomes

Employability	**Skill Development**	**Entrepreneurship**
✓	✓	✓

Course Objectives:

1. To develop an understanding of basic concept of mechanics, and engineering materials for process and textile equipment.
2. To introduce thermodynamics, refrigeration, and heat transfer principles for process heating and cooling applications.
3. To impart knowledge of fluid mechanics, pumps, and compressors used in process and bioprocess fluid systems.
4. To familiarize students with machine elements and fluid power systems used in mechanical transmission applications.
5. To develop an understanding of manufacturing processes for chemical, biomedical, and textile equipment.

Course Content

Unit	**Description**	**Weightage**
1.	**Basic Mechanics and Engineering Materials:** Forces, resultant and equilibrium; stress, strain, Hooke's law and factor of safety; properties and applications of metals, polymers, rubber, ceramics, composites and biomaterials. Applications: materials and structures in chemical, bioprocess and textile equipment.	20%
2.	**Thermodynamics and Heat Transfer:** Thermodynamic system, heat, work and first law; vapour compression refrigeration, refrigeration effect and COP; conduction, convection and radiation. Applications: process heating and cooling, fermenter temperature control, sterilisation, refrigeration and textile drying.	20%

BT Stream

3.	**Pumps, Compressors, and Fluid Systems:** Fluid properties; Bernoulli's equation; centrifugal & reciprocating pumps; priming and pump selection; piston air compressor. Applications: process fluid transport, bioprocess aeration in fermenters, dye-liquor circulation and compressed-air systems.	20%
4.	**Machine Elements:** Fasteners, bearings, shafts, couplings, gears and gear ratio; belt and chain drives; basics of hydraulic and pneumatic systems. Applications: mechanical drives and transmission systems in mixers, centrifuges and textile machinery.	20%
5.	**Chemical, Biomedical Equipment, and Emerging Systems:** Properties and stress–strain behaviour of engineering materials; fundamentals of casting, hot and cold working, and powder metallurgy. Applications: fabrication of bioreactors and process equipment, chemical plant components, porous filters, and textile machinery parts.	20%

Teaching-Learning Methodology

Taught using interactive lectures, chemical equipment videos, and biomedical instrument demonstrations. Every concept is directly linked to textile machinery or biomedical equipment. Case studies, demonstrations, and laboratory exposure will relate concepts to chemical, bioprocess, biomedical, and textile applications.

Course Outcomes (COs)

COs	Course Outcome Statement	KL
CO1	Explain the concept of mechanics, stress–strain, and material properties to process and textile equipment.	K2
CO2	Understand thermodynamic, refrigeration, and heat transfer principles in process thermal systems.	K2
CO3	Explain the concept of fluid principles, pumps, and compressors for process and bioprocess applications.	K2
CO4	Illustrate machine elements and fluid power principles in mechanical transmission systems.	K2
CO5	Understand material properties and manufacturing processes to chemical, biomedical, and textile equipment.	K2

BT Stream

CO–PO Articulation Matrix

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	1	1	1	–	–	–	–	–	–	1
CO2	3	2	2	1	1	–	1	–	–	–	–	1
CO3	3	3	2	2	2	1	1	–	–	–	–	1
CO4	3	2	2	2	2	–	–	–	–	–	–	1
CO5	3	2	2	2	2	1	2	–	–	–	–	2

Evaluation Pattern

Sr. No.	Details of Evaluation	Weightage
1.	Internal Assessment – Class Tests (CT) + Teacher's Assessment (Assignment / Quiz / Attendance)	30%
2.	End Semester Theory Examination	70%

Suggested Textbooks / Reference Books

1. Elements of Mechanical Engineering – R.K. Rajput (S. Chand).
2. Basic Refrigeration and Air Conditioning – P. N. Anantha Narayanan (McGraw Hill Publication).
3. Biomedical Instrumentation (Mechanical Aspects) – R.S. Khandpur (TMH Publication).
4. Chemical Engineering (Vol. 1) – Coulson & Richardson (Butterworth-Heinemann Publication).
5. Principles of Textile Testing – J. E. Booth (CBS Publishers).

Online Resources

1. NPTEL – Introduction to Mechanical Engineering, IIT Bombay

Additional References

Relevant articles, videos, and resources on:

1. Electric Tractors and Autonomous Farm Vehicles
2. Cold Chain Engineering: Design and Management for Biochemical and Textile Products

Back to Course Structure & Evaluation Scheme

BT Stream

Course Code	**AAS151/AAS251**
Title of the Course	**Applied Physics Lab**
Total Credits of the Course	01
NCrF Hours	30

Course Objectives

1. To determine the wavelength of monochromatic light, polychromatic light and specific rotation optical instruments.
2. To determine Planck's constant using photoelectric effect, study of attenuation in optical fibre and verification of Stefan's law.
3. To determine the energy band gap and Hall coefficient of semiconductor materials.
4. To investigate the electrical, magnetic and thermal properties of materials and to verify respective fundamental laws.
5. To verify discrete nature of atomic energy transitions by Franck-Hertz experiment and study the switching characteristics of fundamental logic gates.

List of Experiments (Any 10 experiments relevant to BT stream)

S. No.	Title of Practical	Description/Objective
1.	Diffraction grating	To determine the wavelength of different spectral lines of mercury light using plane transmission grating.
2.	He -Ne Laser	To determine the wavelength of He-Ne laser light using single slit diffraction.
3.	Polarimeter	To determine the specific rotation of cane sugar solution using polarimeter.
4.	Nodal Slide	To determine the focal length of the combination of two lenses separated by a distance and verify the formula for the focal length of combination of lenses.
5.	Planck's constant	To determine Planck's constant using photoelectric effect.
6.	Optical Fiber	To measure attenuation in an optical fiber.
7.	Stefan's Law	To verify Stefan's law by electric method.
8.	Energy Band Gap	To determine the energy band gap of a given semiconductor material.
9.	Hall Effect	To study Hall effect and determine Hall coefficient, carrier density and mobility of a given semiconductor material using Hall effect setup.

BT Stream

10.	Variation of Magnetic Field	To determine the variation of magnetic field with the distance along the axis of a current carrying coil and estimate the radius of the coil.
11.	B-H Curve	To draw hysteresis curve (B-H curve) of a specimen in the form of a transformer and to determine its hysteresis loss.
12.	Carey Foster's Bridge	To determine resistance per unit length and specific resistance of a given resistance using Carey Foster's Bridge.
13.	Electrochemical Equivalent	To determine the electrochemical equivalent (ECE) of copper.
14.	Coefficient of Viscosity	To determine the coefficient of viscosity of a given liquid.
15.	Surface Tension	To measure surface tension of a liquid (e.g., water, ethanol, and surfactant solutions) at room temperature using capillary rise or stalagmometer.
16.	Compound Pendulum	To determine the value of acceleration due to gravity (g) using compound pendulum.
17.	Moment of Inertia	To determine the moment of inertia of a flywheel about its own axis of rotation.
18.	Young's Modulus	To determine the Young's Modulus of a material by Searle's apparatus.
19.	Franck-Hertz Experiment	To determine the first excitation potential of a gas (e.g. Argon) by Franck-Hertz experiment and verify discrete nature of atomic energy transitions.
20.	Solar Cell	To study the characteristics of a solar cell.
21.	Characteristics of Logic Gates	To study the switching characteristics of fundamental logic gates constructed using transistors (BJT/MOSFET), measure propagation delay and power dissipation.

BT Stream

Equipment / Software Required

1. Experimental set-up, inbuilt assembly or specific experiment related apparatus and manuals etc.
2. Light sources, multimeters, rheostat, keys and other required materials.

Teaching-Learning Methodology

Each laboratory session begins with a brief pre-lab discussion connecting the experiment to the corresponding objectives. Students write an aim, apparatus list, procedure, observation table, calculations, result, and inference. Post-lab discussion sessions relate experimental

findings to real-world engineering applications. Students maintain a laboratory record book assessed for viva, record, and observation components.

Course Outcomes (COs)

COs	Course Outcome Statement	KL
CO1	Determine the wavelength of monochromatic light, polychromatic light and specific rotation optical instruments	K3
CO2	Determine Planck's constant using photoelectric effect, study of attenuation in optical fibre and verification of Stefan's law.	K3
CO3	Determine the energy band gap and Hall coefficient of semiconductor materials.	K3
CO4	Investigate the electrical, magnetic and thermal properties of materials and to verify respective fundamental laws.	K3
CO5	Verify discrete nature of atomic energy transitions by Franck-Hertz experiment and study the switching characteristics of fundamental logic gates.	K3

CO–PO Articulation Matrix

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	1	3	3	–	–	–	1	–	–	1
CO2	3	3	1	3	3	–	1	–	1	–	–	1
CO3	3	3	2	3	3	–	–	–	1	–	–	1
CO4	3	3	2	3	3	1	2	–	1	–	–	1
CO5	3	3	2	3	3	–	–	–	1	–	–	2

Evaluation Pattern

Sr. No.	Component	Marks	Weightage
1.	Day-to-day performance (Observation, Procedure, Conduct)	30	30%
2.	Laboratory Record / File	20	20%
3.	End Semester Practical Examination (Experiment + Viva)	50	50%

BT Stream

Safety and General Instructions

1. Students must follow all laboratory safety guidelines and wear appropriate personal protective equipment (PPE).
2. Electrical experiments must be performed only after instructor approval and under supervision.
3. Handle optical instruments, lasers, and high-voltage equipment with extreme care.
4. Chemicals and solvents (where applicable) must be handled as per MSDS guidelines.
5. Equipment must be switched off and lab bench cleaned after every session.

Suggested Textbooks / Reference Books

1. Practical Physics- K. K. Dey & B. N. Dutta (Kalyani Publishers New Delhi
2. Engineering Physics Practical- S K Gupta (Krishna Prakashan Meerut)
3. Engineering Physics: Theory and Practical- Katiyar and Pandey (Wiley India)
4. Engineering Physics Theory and Experiments-Srivastava and Yadav (New Age International)

Back to Course Structure & Evaluation Scheme

BT Stream

Course Code	**AAS152F/ AAS252F**
Title of the Course	Applied Chemistry for Biomolecules and Bioprocess Systems Lab
Course Category	Basic Science Course (BSC)
Total Credits	01
NCrF Hours	30

Main Focus of the Course Outcomes

Employability	**Skill Development**	**Entrepreneurship**
✓	✓	✓

Course Objectives:

1. To identify and quantify biomolecules — carbohydrates, proteins, lipids, and nucleic acids
2. To study enzyme kinetics and determine Km and Vmax for an enzyme-substrate system
3. To analyse fermentation broth and study metabolic products using chemical methods
4. To characterise biosensor responses and bioprocess media quality parameters
5. To apply green chemistry principles in biosynthesis and bioprocess systems

List of Experiments *(Any 10 Experiments)*

Exp.	**Theory Unit**	**Title**	**Description / Objective**
1	Unit I	Qualitative Tests for Carbohydrates	Perform Molisch's, Benedict's, Fehling's, Barfoed's, and Seliwanoff's tests on given sugar solutions; identify reducing sugars, non-reducing sugars, monosaccharides, and polysaccharides
2	Unit I	Quantitative Estimation of Glucose by DNS Method	Estimate glucose concentration in unknown sample using 3,5-dinitrosalicylic acid (DNS) reagent; plot calibration curve; determine unknown concentration by UV-Vis spectrophotometry

BT Stream

3	Unit I	Estimation of Protein by Bradford / Lowry Method	Estimate protein concentration in a given sample using Bradford or Lowry method; plot BSA standard curve; determine protein content by colorimetry; relate to bioprocess broth analysis
4	Unit II	Effect of Temperature on Enzyme Activity	Study the effect of temperature (20°C–80°C) on amylase or urease activity using starch-iodine or Nessler's method; determine optimum temperature; relate to bioreactor operating conditions
5	Unit II	Effect of pH on Enzyme Activity	Study effect of pH on amylase or protease activity using buffer solutions; determine optimum pH; relate to enzyme immobilisation and bioprocess pH control
6	Unit III	Estimation of Ethanol in Fermentation Broth	Determine ethanol content in fermentation broth by dichromate oxidation method or ebulliometer; relate to bioethanol fermentation process monitoring and yield calculation
7	Unit III	Estimation of Total Reducing Sugars in Fermentation Medium	Determine total reducing sugar content in fermentation medium before and after fermentation using DNS method; calculate substrate utilisation and fermentation efficiency
8	Unit IV	Biosensor Response – Glucose Biosensor Calibration	Calibrate a commercial glucose biosensor (glucometer) against standard glucose solutions; plot calibration curve; determine sensitivity and linearity; relate to enzyme biosensor working principle
9	Unit IV	Water Activity and pH of Fermentation Media	Measure pH and estimate water activity of different fermentation media; relate to microbial growth kinetics and substrate availability in bioprocess systems
10	Unit V	Green Synthesis of Silver Nanoparticles Using Plant Extract	Synthesise silver nanoparticles using neem or tulsi leaf extract as reducing and capping agent; confirm by UV-Vis spectroscopy (plasmon peak ~420 nm); relate to biogenic synthesis and green nanotechnology

BT Stream

11	Unit V	Estimation of Lipids by Soxhlet Extraction	Determine lipid content of a given biological sample (oilseed, yeast) by Soxhlet extraction using petroleum ether; calculate % lipid; relate to biodiesel feedstock evaluation
12	Unit I	Qualitative Tests for Amino Acids and Proteins	Perform Biuret, Ninhydrin, Xanthoproteic, and Millon's tests on amino acid and protein solutions; identify protein types; relate to downstream protein quality analysis

Equipment / Chemicals Required

1. UV-Vis spectrophotometer
2. Analytical balance and centrifuge
3. Water bath / thermostatic bath
4. pH meter and buffer solutions
5. DNS reagent, Bradford/Lowry reagent kits
6. Soxhlet extraction apparatus
7. Fermentation broth samples (prepared in advance)
8. Commercial glucose biosensor (glucometer)
9. Reflux condenser and COD glassware
10. Microwave / water bath for nanoparticle synthesis

Teaching-Learning Methodology

Each laboratory session begins with a 10-minute pre-lab discussion connecting the experiment to the corresponding theory unit. Students record aim, principle, apparatus, chemicals, procedure, observation table, calculations, result, and inference in the laboratory record book. Post-lab discussion relates experimental findings to real-world engineering applications in the respective stream. Students maintain a laboratory record book assessed for viva, record, and day-to-day performance.

Course Outcomes (COs)

COs	Course Outcome Statement	KL
CO1	Perform qualitative and quantitative analysis of carbohydrates, proteins, and lipids	K3
CO2	Study effect of temperature and pH on enzyme activity and determine optimum conditions	K3

BT Stream

CO3	Estimate ethanol and reducing sugars in fermentation broth and calculate fermentation efficiency	K3
CO4	Calibrate a glucose biosensor and measure fermentation media quality parameters	K3
CO5	Synthesise silver nanoparticles by green route and determine lipid content by Soxhlet extraction	K3

CO-PO Articulation Matrix

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	1	3	2	–	–	–	1	–	–	1
CO2	3	3	2	3	2	–	–	–	1	–	–	1
CO3	3	3	2	3	2	1	2	1	1	–	–	1
CO4	3	2	3	3	2	1	2	1	1	–	–	1
CO5	3	2	2	3	2	2	3	2	2	1	–	2

Evaluation Pattern

Sr.	Component	Marks	Weightage
1.	Day-to-day performance (Observation, Procedure, Conduct)	20	20%
2.	Laboratory Record / File	30	30%
3.	End Semester Practical Examination (Experiment + Viva Voce)	50	50%

Suggested Textbooks / Reference Books

1. Engineering Chemistry – Jain and Jain
2. Biochemistry – Lehninger, Nelson, and Cox
3. Bioprocess Engineering Principles – Pauline M. Doran

Safety and General Instructions

1. Students must wear lab coat and safety goggles at all times in the chemistry laboratory
2. Handle all acids, alkalis, and oxidising agents with extreme care; use fume hood for volatile chemicals

BT Stream

3. Dispose of chemical waste in designated containers as per lab safety guidelines
4. Do not pipette by mouth; use rubber bulb or mechanical pipette fillers
5. Wash hands thoroughly after every experiment; report any spillage or accident immediately

Back to Course Structure & Evaluation Scheme

BT Stream

Course Code	**AEE151C / AEE251C**
Title of the Course	Electrical Engineering Lab
Course Category	Engineering Science Course (ESC)
Total Credits	01
NCrF Hours	30

Main Focus of the Course Outcomes

Employability	**Skill Development**	**Entrepreneurship**
✓	✓	✓

Course Objectives:

1. Verify Ohm's law and Kirchhoff's laws through simple DC circuit experiments relevant to lab and field instruments
2. Measure AC power and power factor in equipment used in agriculture and biotechnology
3. Study the fundamentals of electrical measurements and operation of measuring instruments.
4. Study transformer efficiency and motor starting for agri-pump and biomedical equipment selection
5. Demonstrate electrical safety, working of Fluorescent Lamp, electrical safety and protection equipment and energy measurement as core practical skills for farm and lab environment.

List of Experiments *(Any 10 Experiments)*

Exp.	**Theory Unit**	**Title**	**Objective / Description with Application**
1	Unit V	Electrical Safety Practices in the Laboratory	Demonstrate safe handling of electrical equipment, proper use of insulated tools, earthing, and emergency shutdown procedures. Application: Safe operation of electrical systems in laboratories and workplace
2	Unit I	Verification of KVL and KCL	Verify Kirchhoff's Voltage Law in a series circuit and Kirchhoff's Current Law in a parallel circuit.

BT Stream

3	Unit I	Thevenin's Theorem Verification	Find Thevenin's equivalent of a DC network; verify with different load resistors.
4	Unit II	Power and Power Factor Measurement	Connect a single-phase inductive load (choke / single-phase motor); measure voltage, current, and active power using a wattmeter; compute power factor.
5	Unit II	Power Factor Improvement Using Capacitor	Improve power factor of an inductive load by adding capacitors in steps; record reduction in line current.
6	Unit III	Measurement of DC Voltage and Current Using PMMC Instruments	Measure DC voltage and current using PMMC voltmeter and ammeter; verify the readings with a digital multimeter. Application: Battery systems, DC power supplies, and control circuits.
7	Unit III	Measurement of AC Voltage and Current Using Moving Iron (MI) Instruments	Measure AC voltage and current using MI voltmeter and ammeter; compare the readings with a digital multimeter. Application: Electrical distribution panels and industrial AC circuit
8	Unit III	Measurement of Single-Phase Power Using a Dynamometer Wattmeter	Measure active power in a single-phase circuit using a dynamometer wattmeter and verify the result by calculation. Application: Power measurement of electrical appliances and industrial loads.
9	Unit IV	Single-Phase Transformer – Load Test	Perform a load test on a step-down transformer; measure primary and secondary voltages and currents at various loads; compute efficiency and
10	Unit IV	Starting of a Three-Phase Induction Motor Using DOL Starter	Start a three-phase induction motor using a Direct-On-Line (DOL) starter and observe its operation and rotation. Application: Pumps, fans, blowers, and conveyors used in industries.
11	Unit V	Working of Fluorescent Lamp Circuit	Connect and observe the operation of a fluorescent lamp using a choke and starter; identify the function of each component.

BT Stream

			Application: Lighting systems in schools, offices, and industries.
12	Unit V	Study of Fuse and MCB Operation	Observe the operation of a fuse and Miniature Circuit Breaker (MCB) under overload and short-circuit conditions (demonstration). Application: Protection of domestic and industrial electrical installation

Equipment / Instruments Required

1. DC Regulated Power Supply and Digital Multimeters
2. Resistor Kit and Rheostat
3. Single-Phase AC Supply with Variac
4. AC/DC Ammeters, Voltmeters and Wattmeters
5. Choke Coil (Inductive Load) and Capacitor Bank
6. Three-Phase Supply Panel with Balanced Star Resistive Load
7. Two Single-Phase Wattmeters (for Three-Phase Power Measurement)
8. Single-Phase Transformer with Load Panel
9. Three-Phase Squirrel Cage Induction Motor with DOL Starter
10. PMMC and Moving Iron (MI) Instruments
11. Dynamometer Type Wattmeter
12. Fluorescent Lamp with Choke, Starter and Lamp Holder
13. MCB, Fuse Demonstration Board and Earth Continuity Tester
14. Connecting Wires, Switches and Terminal Blocks
15. Safety Equipment (Insulated Gloves, Safety Goggles, Fire Extinguisher and First Aid Kit)

Teaching-Learning Methodology

Each 2-hour session begins with a 10-minute pre-lab briefing linking the experiment to its theory unit and a real engineering application. Students record aim, circuit diagram, apparatus, procedure, observation table, sample calculations, result, and inference in their lab record book. A 10-minute post-lab discussion relates the experiment to emerging technologies in the stream. Continuous assessment of observation quality and conduct is maintained.

Course Outcomes (COs)

COs	Course Outcome Statement	KL
CO1	Verify Ohm's law, KVL, KCL, and Thevenin's theorem in DC circuits relevant to agri and biomedical instruments	K3

BT Stream

CO2	Measure AC power and power factor and demonstrate PF improvement for agri and biomedical loads	K3
CO3	Study the fundamentals of electrical measurements and operation of measuring instruments	K2
CO4	Evaluate transformer efficiency and study motor starting for laboratory setup and industrial applications.	K3
CO5	Demonstrate electrical safety, working of Fluorescent Lamp, electrical safety and protection equipment.	K2

CO-PO Articulation Matrix

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	1	3	3	–	–	–	1	–	–	1
CO2	3	3	2	3	3	–	1	–	1	–	–	1
CO3	3	3	2	3	3	–	1	–	1	–	–	1
CO4	3	2	2	3	3	–	1		1	–	–	1
CO5	3	2	1	2	3	2	2	3	1	–	–	1

Evaluation Pattern

Sr.	Component	Marks	Weightage
1.	Day-to-day performance (Observation, Circuit, Conduct)	20	20%
2.	Laboratory Record / File	30	30%
3.	End Semester Practical Examination (Experiment + Viva Voce)	50	50%

Safety Instructions

1. No circuit should be energised without the instructor's approval
2. Always switch OFF supply before making or changing any connection
3. Three-phase experiments must be performed under direct instructor supervision
4. Never exceed the rated range of any measuring instrument
5. Switch off all equipment and tidy the lab bench after every session
6. Study transformer efficiency and motor starting for agri-pump and biomedical equipment selection

Back to Course Structure & Evaluation Scheme

BT Stream

Course Code	**AEC151C / AEC251C**
Title of the Course	Electronics Engineering Lab
Course Category	Engineering Science Course (ESC)
Total Credits	01
NCrF Hours	30

Main Focus of the Course Outcomes

Employability	**Skill Development**	**Entrepreneurship**
✓	✓	✓

Course Objectives:

1. Set up a diode rectifier and Zener voltage regulator for biomedical and agri instrument supply
2. Use a transistor switch to drive a relay or solenoid valve for agri or lab automation
3. Calibrate temperature, pH, and soil moisture sensors and condition their signals with an op-amp
4. Verify digital logic gates and understand microcontroller-based sensor data acquisition
5. Demonstrate a solar PV circuit, an IoT soil monitoring node, or a simple biomedical sensor circuit

List of Experiments *(Any 10 Experiments)*

Exp.	**Theory Unit**	**Title**	**Objective / Description with Application**
1	Unit I	Diode Characteristics and Bridge Rectifier	Plot V-I characteristic of a silicon diode; set up a full-wave bridge rectifier with filter capacitor; measure DC output. Application: DC supply for a portable pH meter or field soil sensor unit.
2	Unit I	Zener Diode Voltage Regulator	Set up Zener regulator; vary supply voltage and load; measure regulated output. Application: regulated 5V supply for an Arduino-based irrigation controller or biomedical instrument.

B. Stream

3	Unit I	LED and Photodiode Characteristics	Measure LED forward voltage and current at different current levels; observe LED brightness; measure photodiode reverse current vs. illumination. Application: LED indicator in diagnostic equipment; photodiode in pulse oximeter (SpO2) light detection.
4	Unit II	BJT as a Switch – Relay Driver for Solenoid Valve	Use NPN BJT to switch a relay; connect relay to control a solenoid valve (or motor) ON/OFF using a small control signal. Application: automated irrigation valve control triggered by a soil moisture sensor reading.
5	Unit III	Thermistor Sensor Calibration	Measure resistance of a thermistor at different temperatures (room to 80°C); plot R-T curve; convert to temperature using calibration equation. Application: temperature monitoring in fermenters, incubators, and cold storage for agri-produce.
6	Unit III	Op-Amp Signal Amplifier for Sensor Output	Connect thermistor (in bridge) to non-inverting op-amp amplifier (IC 741); measure amplified output voltage for different temperatures. Application: amplifying the tiny output of a biosensor or thermocouple for display or ADC input.
7	Unit III	Op-Amp Comparator – Temperature or Light Alarm	Set up op-amp comparator with sensor input and adjustable reference; drive an LED or buzzer when threshold is crossed. Application: over-temperature alarm in biomedical autoclave; under-light alarm in greenhouse; frost alarm in farm.

BT Stream

8	Unit III	Soil Moisture Sensor – Calibration and Signal Conditioning	Measure output voltage of a capacitive or resistive soil moisture sensor at dry, moist, and saturated soil conditions; amplify output with op-amp if needed. Application: drip irrigation automation — turn pump ON when soil moisture drops below threshold.
9	Unit IV	Logic Gate Truth Table Verification	Verify AND, OR, NOT, NAND gate truth tables using a logic trainer; implement simple control logic: Pump ON = Soil_Dry AND (NOT Tank_Empty). Application: basic automated irrigation logic; digital control in biomedical instrument.
10	Unit V	Solar Cell – I-V Curve and Battery Charging Circuit	Plot I-V curve of a small solar cell; connect to a 12V battery via a simple charge controller; measure charging current. Application: solar-powered soil sensor node; off-grid poultry farm lighting; portable biomedical device charging in rural areas.
11	Unit V	IoT Soil / Environment Sensor Node (Arduino/NodeMCU)	Connect soil moisture and temperature sensors to an Arduino/NodeMCU; read sensor values on serial monitor; display on LCD; (if Wi-Fi) send data to ThingSpeak/Blynk. Application: precision irrigation monitoring; smart greenhouse; remote biomedical patient environment monitoring.
12	Unit V	Pulse Oximeter / Heart Rate Sensor Demo (Biomedical)	Connect an IR + red LED pulse sensor (MAX30100 or similar) to an Arduino; display SpO2 and heart rate on serial monitor or LCD. Application: understanding the electronics inside a clinical

BT Stream

			pulse oximeter; wearable health monitor concept.

Equipment / Instruments Required

1. DC regulated power supply and digital multimeters
2. Breadboard and component kit (diodes, Zener, NPN BJT, IC 741, LED, photodiode)
3. Thermistor (NTC) with temperature bath and reference thermometer
4. LDR and soil moisture sensor (resistive or capacitive)
5. Relay (5V), buzzer, small solenoid valve or DC motor
6. Digital logic trainer (ICs 7408, 7432, 7404, 7400)
7. Solar cell module (5W), 12V battery, simple charge controller
8. Arduino / NodeMCU + LCD (16x2) + DHT11 + soil moisture sensor + USB cable
9. MAX30100 / MAX30102 pulse oximeter sensor module + Arduino (for Exp. 12)
10. Laptop with Tinkercad / Falstad (browser-based, free)

Teaching-Learning Methodology

Each 2-hour session begins with a 10-minute pre-lab discussion linking the experiment to its theory concept and a real product or industry application. Students build circuits on a breadboard, take measurements, and record observations, calculations, result, and inference. Post-lab discussion highlights how the experiment relates to a real device (phone charger, PLC output, biosensor, IoT node). Simulation (Falstad/Tinkercad) is used alongside hardware wherever possible.

Course Outcomes (COs)

COs	Course Outcome Statement	KL
CO1	Set up diode rectifier, Zener regulator, and LED/photodiode circuits for BT/AG instrument supply	K3
CO2	Use BJT switch to drive relay and control solenoid valve or actuator in agri or biomedical circuit	K3
CO3	Calibrate temperature and soil moisture sensors and apply op-amp amplification and comparator alarm	K3
CO4	Verify logic gate truth tables and implement simple automated control logic for BT/AG systems	K3

BT Stream

CO5	Demonstrate solar PV charging, IoT sensor monitoring, and biomedical sensor interfacing	K3

CO-PO Articulation Matrix

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2	3	3	–	–	–	1	–	–	1
CO2	3	3	2	3	3	–	–	–	1	–	–	1
CO3	3	3	2	3	3	–	–	–	1	–	–	1
CO4	3	2	2	3	3	–	–	–	1	–	–	1
CO5	3	3	3	3	3	1	1	–	2	1	–	2

Evaluation Pattern

Sr.	Component	Marks	Weightage
1.	Day-to-day performance (Circuit building, Observation, Conduct)	20	20%
2.	Laboratory Record / File	30	30%
3.	End Semester Practical Examination (Experiment + Viva Voce)	50	50%

Safety Instructions

1. Always build the circuit with power OFF; energise only after instructor check
2. Do not apply voltage exceeding IC or component ratings — check datasheet
3. Handle CRO probes and signal generator outputs carefully
4. Do not connect 230V AC load directly — use relay output with instructor supervision only
5. Switch off all instruments and tidy the breadboard and bench after every session

Back to Course Structure & Evaluation Scheme

BT Stream

Course Code	**ACE151G / ACE251G**
Title of the Course	Computer Aided Engineering Drawing Lab
Course Category	Engineering Science Course (ESC)
Total Credits	02
NCrF Hours	45

Main Focus of the Course Outcomes

Employability	**Skill Development**	**Entrepreneurship**
✓	✓	✓

Course Objectives

1. To familiarise the student with the principles, conventions and standards of engineering drawing, so that a drawing can be read and prepared without ambiguity.
2. To develop the ability to visualise a three-dimensional object and to represent it through orthographic, sectional and pictorial views.
3. To develop working competence in a CAD package for two-dimensional drafting and for basic three-dimensional modelling.
4. To develop the ability to prepare drawings of the components, systems and layouts that occur in the Biotechnology Stream (BT), rather than generic examples.
5. To develop the ability to prepare equipment, facility layout and device drawings relevant to bioprocess and biomedical practice, and to introduce 3D modelling and printing as used in the bio industry.

List of Experiments / Exercises		
Exp. No.	**Title and Brief Description**	**Hours**
1	**Drawing standards, sheet layout and the CAD environment** Sheet sizes, borders, title block, lines, lettering and dimensioning as per BIS conventions (SP 46 : 2003 (BIS) Engineering Drawing Practice for Schools and Colleges; IS 10711 (sheet sizes and layout); IS 9609 (lettering); IS 10714 (line conventions); IS 11669 (preparation of general engineering drawings); ISO 128 series.). Familiarisation with the CAD interface: coordinate entry, units, limits, object snap, layers, linetypes and file management.	3

	Students set up a standard A3/A4 template with a title block that will be reused throughout the semester.	
2	**Geometric constructions and engineering curves** Construction of polygons, tangents, plain and diagonal scales; conic sections (ellipse, parabola, hyperbola) and one roulette (cycloid or involute) using CAD draw and modify commands (LINE, ARC, OFFSET, TRIM, ARRAY, MIRROR). Stream context: the torispherical dished end of a bioreactor vessel, the helix of a coil/spiral membrane module, and the parabolic profile of a vortex in an unbaffled vessel.	3
3	**Orthographic projection of points and straight lines** Projections of points in the four quadrants; projections of lines inclined to one and both reference planes; true length and true inclination by rotation/trapezoidal method. Principal planes handled on separate CAD layers.	3
4	**Orthographic projection of planes and simple solids** First angle projection of planes and of prisms, pyramids, cylinders and cones resting in given positions; front, top and side views. Third angle convention introduced for comparison, since it is common in imported drawings.	3
5	**Sections of solids and true shape of section** Solids cut by section planes perpendicular to one reference plane and inclined to the other; sectional views, cutting plane line, hatching conventions and true shape of the section obtained by auxiliary projection.	3
6	**Development of lateral surfaces** Parallel line and radial line development of cube, prism, cylinder, cone and truncated solids. Stream context: development of a conical vessel bottom, a cyclone/separator body and a sheet metal instrument housing.	3
7	**Isometric projection and reading of orthographic views** Isometric scale, isometric axes and isometric drawing aids in CAD; conversion of given orthographic views into an isometric view, and the reverse exercise of reading a pictorial view and producing orthographic views. Missing line and missing view problems for visualisation practice.	3
8	**Drawing of a bioprocess equipment component** Orthographic and sectional drawing of a stirred tank bioreactor: vessel with dished ends, jacket, baffles, Rushton turbine impeller with shaft and seal, sparger and nozzles. Key ratios (H/D, D_i/D, baffle width) are marked so	4

BT Stream

	that the drawing carries process meaning and not merely geometry.	
9	**Process/utility flow diagram and cleanroom laboratory layout** Flow diagram of a simple upstream–downstream sequence (media preparation, sterilisation, fermentation, centrifugation/filtration, chromatography) with equipment symbols and stream labels; and a layout drawing of a laboratory or a small cleanroom suite showing airlocks, material and personnel flow paths, and pressure cascade, with a note on why the flows must not cross.	4
10	**3D modelling of a bio component or a biomedical device part** Solid model of an impeller, a filter housing, or for the biomedical branch a simple device component such as a prosthetic/orthotic part, a scaffold with a controlled pore pattern or an instrument casing; automatic generation of orthographic and sectional views from the model.	4
11	**Sheet composition, plotting and export for 3D printing** Composition of a complete sheet with title block, sectional views and dimensioning for the component of Experiment 10; plotting to scale; export to PDF/DXF and to STL, with slicing parameters examined and the effect of orientation and layer thickness on the printed part discussed.	3
12	**Mini project with an introduction to AI assisted bio design** In groups of two or three, students prepare a complete drawing set — equipment or device drawing with a 3D model, plus the associated flow diagram or layout. The session includes a demonstration of image derived (CT/MRI) patient specific modelling, bioprinting workflows, lab on chip/microfluidic layout, and AI assisted design and shape optimisation in biomedical applications. Evaluation by presentation and viva voce.	6
	Internal assessment, viva voce and end semester practical examination	**03**
	Total	**45**

Teaching–Learning Methodology
Each weekly session begins with a short tutorial (1 hour) on the drawing principle or standard involved, followed by two hours of hands-on CAD work at the terminal. Experiments 1 to 7 build the common graphics

BT Stream

foundation. Experiments 8 to 12 are specific to the Biotechnology Stream (BT) and are conducted by, or in consultation with, the parent department. Every student maintains a CAD drawing portfolio (soft copy plus a plotted file) which is submitted for internal assessment. Sheets are checked for correctness of projection and for conformity to standards a geometrically correct but non-conforming drawing is returned for correction. The mini project of Experiment 12 is done in groups of two or three and is defended individually in a viva voce.

Course Outcomes (COs)

CO	Course Outcome Statement (on successful completion of the course, the student will be able to:)	KL
CO1	Apply engineering drawing standards and conventions (BIS/ISO) while setting up and dimensioning a drawing.	3
CO2	Construct orthographic and sectional views of points, lines, planes and solids using a CAD package.	3
CO3	Develop surfaces and isometric views of solids, read given views to visualise the object, and apply this to draw equipment, flow diagrams and facility layouts for bioprocess and biomedical practice.	3
CO4	Create three-dimensional models in CAD, derive two-dimensional views from them, and exchange drawing data in standard formats.	3
CO5	Compose, plot and defend a complete stream specific drawing set through a mini project, using contemporary CAD practice and emerging parametric/AI assisted tools.	6

CO – PO Articulation Matrix

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2	2	2				1	1		1
CO2	3	3	2	2	2				1	1		1
CO3	2	2	3	2	2		1		2	1	1	2
CO4	2	1	3	2	3				1	1		2
CO5	2	2	3	2	3	1	1	1	3	3	2	3

BT Stream

Evaluation Pattern		
Sr. No.	**Details of the Evaluation**	**Weightage**
1	Internal Assessment (PS): continuous evaluation of experiment performance, the drawing portfolio/lab record, mini project and viva voce, and attendance	50 %
2	End Semester Practical Examination (PE): CAD exercise assigned on the spot (one common and one stream specific component), drawing record and viva voce	50 %

Suggested Reference (Latest Edition)	
1.	N. D. Bhatt and V. M. Panchal, Engineering Drawing, Charotar Publishing House, Anand (latest edition).
2.	P. F. Stanbury, A. Whitaker and S. J. Hall, Principles of Fermentation Technology, Butterworth Heinemann/Elsevier (latest edition).
3.	M. L. Shuler and F. Kargi, Bioprocess Engineering: Basic Concepts, Pearson Education (latest edition).
4.	R. S. Khandpur, Handbook of Biomedical Instrumentation, McGraw Hill Education (latest edition).
5.	K. Venugopal, Engineering Drawing and Graphics + AutoCAD, New Age International Publishers (latest edition).
6.	ISO 10628, Diagrams for the Chemical and Petrochemical Industry (for flow diagram conventions), and SP 46 : 2003, Engineering Drawing Practice for Schools and Colleges, Bureau of Indian Standards.
7.	Pandey, S. K. (2001). *Learn AutoCAD: In a Easy Way*. S.K. Kataria & Sons. ISBN: 978-81-85749-78-5.
Online / e Resources	
1.	NPTEL/SWAYAM: Engineering Drawing and Computer Graphics, Prof. Rajaram Lakkaraju, IIT Kharagpur
2.	NPTEL/SWAYAM: Engineering Graphics and Design, Prof. Naresh V. Datla and Prof. S. R. Kale, IIT Delhi
3.	FreeCAD / Autodesk Fusion official documentation and tutorials.
4.	3D Slicer open-source documentation, https://slicer.org for the demonstration on image derived patient specific models.

Back to Course Structure & Evaluation Scheme

BT Stream

Course Context: This is a Project-Based Learning (PBL) course aligned with the AICTE model curriculum on Design Thinking & Innovation. Students work in teams through the complete Design Thinking cycle — Empathize, Define, Ideate, Prototype, Test — to solve a real-world problem. The emphasis is on doing, not lecturing: every session is a workshop or studio activity. The course builds creativity, teamwork, problem-solving, and an entrepreneurial mindset in first-year students of all branches.

Course Code	**AID151/AID251**
Title of the Course	Innovation & Design Thinking Lab (Project-Based Learning)
Course Category	Skill Development Course (SDC)
Total Credits	02
NCrF Hours	45
Mode	Studio / Workshop — Team-based Project Learning

Main Focus of the Course Outcomes

Employability	**Skill Development**	**Entrepreneurship**
✓	✓	✓

Course Objectives:

1. To introduce students to the Design Thinking process as a human-centred approach to problem solving
2. To develop empathy and the ability to identify real user needs and define meaningful problems
3. To build creative ideation, brainstorming, and lateral-thinking skills
4. To enable students to build low-cost prototypes and test them with users
5. To foster teamwork, communication, and an innovation and entrepreneurial mindset through a hands-on team project

Course Content — Design Thinking Modules (AICTE Aligned)

Module	**Description**	**Hours**

BT Stream

1.	**Module 1 – Introduction to Design Thinking & Innovation:** What is Design Thinking; difference between design thinking and conventional problem solving; the five stages (Empathize, Define, Ideate, Prototype, Test); innovation vs invention; examples of design thinking success stories (Apple, Airbnb, IDEO, Indian startups); creativity and the innovation mindset; introduction to the team project.	6 hrs
2.	**Module 2 – Empathize: Understanding the User:** Human-centred design; observation and user research techniques; conducting interviews and surveys; empathy mapping; identifying user needs, pain points, and latent needs; stakeholder analysis; developing personas. Studio Activity: field observation and user interviews for the chosen problem.	8 hrs
3.	**Module 3 – Define & Ideate:** Synthesising research; framing the problem statement (Point of View); 'How Might We' questions; brainstorming rules and techniques; SCAMPER, mind mapping, lateral thinking; idea selection and prioritisation (impact-effort matrix, dot voting); concept development. Studio Activity: define problem statement and generate/short-list solution ideas.	10 hrs
4.	**Module 4 – Prototype:** Purpose of prototyping; types of prototypes (paper, low-fidelity, working model, digital mock-up); rapid prototyping with low-cost materials; storyboarding; introduction to prototyping tools (cardboard, clay, 3D printing basics, Tinkercad, Figma/Canva for digital prototypes). Studio Activity: build a low-cost prototype of the team's solution.	12 hrs
5.	**Module 5 – Test, Iterate & Pitch:** User testing and feedback collection; iteration and refinement based on feedback; failure as learning; introduction to business model canvas (basic);	9 hrs

BT Stream

	value proposition; storytelling and pitching; preparing and delivering the final project pitch. Studio Activity: test prototype with users, iterate, and present the final pitch.	

Course Outcomes (COs)

COs	Course Outcome Statement	KL
CO1	Explain the design thinking process and apply the innovation mindset to real problems	K2
CO2	Apply empathy tools (interviews, observation, empathy maps) to identify genuine user needs	K3
CO3	Define a clear problem statement and generate creative solutions through structured ideation	K3
CO4	Build a low-cost prototype of a proposed solution using appropriate tools	K3
CO5	Test and iterate the prototype based on user feedback and pitch the solution effectively as a team	K4

CO-PO Articulation Matrix

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2	2	–	1	1	–	1	1	1	–	3
CO2	1	3	2	2	2	2	–	2	2	2	–	2
CO3	2	3	3	2	2	2	1	2	2	2	1	2
CO4	2	2	3	3	3	1	1	1	2	1	2	2
CO5	2	3	3	3	2	2	1	2	3	3	2	3

Evaluation Pattern

Sr.	Component	Marks	Weightage
1.	Internal Assessment (IA) — Stage-wise deliverables (Empathy report, Problem statement, Ideation, Prototype), Team-work, Design journal / portfolio	50	50%
2.	External Assessment (EA) — Final prototype demonstration + Project pitch + Viva Voce (by external examiner)	50	50%

BT Stream

Internal Assessment — Detailed Rubric (50 Marks)

Component	Deliverable	Marks
Empathize Stage	User research + empathy map + personas	10
Define Stage	Problem statement (POV) + How-Might-We	8
Ideate Stage	Brainstorm record + short-listed concept	8
Prototype Stage	Low-cost working prototype / model	12
Design Journal	Team logbook / portfolio of the whole process	7
Team Participation	Individual contribution + collaboration	5

Suggested Project Themes (Teams may choose or propose their own)

1. Campus & Community — waste management, water conservation, accessibility for the differently-abled, canteen queue management
2. Agriculture & Rural — low-cost farming tools, water-saving irrigation, crop storage, farmer information access
3. Health & Wellbeing — elderly care aids, hygiene solutions, mental wellness, first-aid tools
4. Sustainability & Environment — plastic reduction, energy saving, green mobility, air-quality awareness
5. Education & Skilling — learning aids, tools for rural schools, assistive learning for disabilities
6. Smart Living — affordable home automation, safety devices, smart utilities

Suggested Learning Resources

1. AICTE Model Curriculum – Design Thinking (aicte-india.org)
2. The Design of Everyday Things – Don Norman
3. Change by Design – Tim Brown (IDEO)
4. Design Thinking Bootleg – Stanford d.school (dschool.stanford.edu, free)
5. IBM Enterprise Design Thinking – ibm.com/design/thinking (free)
6. NPTEL – Design Thinking: A Primer (IIT Madras)

Back to Course Structure & Evaluation Scheme

BT Stream

Course Code	**AAS153 / AAS253**
Title of the Course	Language Lab
Course Category	Applied Science & Humanities (AS&H)
Total Credits	Non-Credit
NCrF Hours	30

Main Focus of the Course Outcomes

Employability	**Skill Development**	**Entrepreneurship**
✓	✓	

Course Objectives

1. Develop proficiency in Listening, Speaking, Reading and Writing (LSRW) skills for academic, professional and workplace communication.
2. Build confidence in spoken English through interactive, collaborative and technology-enabled learning.
3. Strengthen pronunciation, fluency, vocabulary, grammar and presentation skills using digital language learning tools.
4. Develop professional communication skills required for interviews, group discussions, presentations, meetings and workplace interactions.
5. Enhance employability, critical thinking, leadership, teamwork and ethical communication through experiential learning and AI-assisted language practice

Unit	**Description**
1	**Part I: Interactive Communication Skills Lab** The laboratory shall provide experiential learning through role plays, simulations, presentations, discussions and workplace communication activities. **List of Practical Activities** 1. Self-Introduction, Video Introduction, and SWOT Analysis 2. Professional Networking and Digital Professional Branding (LinkedIn, X, or Professional Blogs) 3. Business Communication: Email, Notice, Agenda, Minutes of Meeting, and Technical Report Writing 4. Group Discussion (Case-based, Panel Discussion, and Turn-taking Techniques) 5. Role Play on Workplace Communication, Professional Etiquette, Negotiation, and Conflict Resolution

BT Stream

	6. Mock Interview (HR, Technical, Behavioral, and Situational Interviews) 7. Extempore Speech, Just A Minute (JAM), and Elevator Pitch 8. Technical and AI-assisted Presentation with Ethical Use of AI 9. Non-verbal Communication: Body Language (Kinesics), Eye Contact, Gestures, Proxemics, Facial Expressions, and Voice Modulation 10. Listening and Phonetics: IPA Basics, Pronunciation, Stress, Intonation, Accent Neutralization, and Audio-Video Comprehension using language learning software 11. Critical Thinking through Case Studies, Debate, Storytelling, and Collaborative Problem-solving 12. Professional Digital Portfolio: Résumé, Cover Letter, Video Portfolio and Career Portfolio
2	**Part II: Computer-Assisted Language Learning** Students shall complete self-paced software-based language* learning modules aligned with **CEFR (Target Level: B2)** to enhance employability and workplace communication. **Suggested Modules** • Grammar and Vocabulary • Listening and Reading Comprehension • Pronunciation, Phonetics, Stress, and Intonation • Spoken English and Fluency • Business Communication and Email Writing • Interview Preparation and Presentation Skills **Suggested Tools** • Praat : https://www.fon.hum.uva.nl/praat/ • BBC Learning English : https://www.bbc.co.uk/learningenglish • Sounds of Speech : http://soundsofspeech.uiowa.edu/ • British Council Learn English : https://learnenglish.britishcouncil.org/ • Forvo : https://forvo.com/ • Mozilla Common Voice : https://commonvoice.mozilla.org/en • British council: English score app (Google play store) ***Note**: The institute may use any software that meets the suggested modules criteria, depending on its availability.

Course Outcomes

COs	Course Outcome Statement	KL
CO1	Demonstrate effective Listening, Speaking, Reading and Writing (LSRW) skills for academic, professional and workplace communication.	K3

BT Stream

CO2	Apply appropriate verbal and non-verbal communication skills in presentations, group discussions, interviews and workplace interactions.	K3
CO3	Use correct pronunciation, grammar, vocabulary and digital language learning tools to improve fluency and communication effectiveness.	K3
CO4	Analyse professional communication situations and prepare effective reports, presentations and formal workplace documents using appropriate communication strategies.	K4
CO5	Demonstrate employability skills, teamwork, leadership, ethical communication and AI-assisted communication practices in professional environments.	K3

CO-PO Articulation Matrix

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	1	1	–	–	2	–	–	–	2	3	–	1
CO2	1	2	1	–	2	1	–	2	3	3	1	1
CO3	1	2	–	–	3	–	–	1	2	3	–	2
CO4	1	3	2	1	2	1	–	2	2	3	1	2
CO5	1	2	2	–	3	2	1	3	3	3	2	3

Evaluation Pattern		
Sr. No.	**Details of the Evaluation**	**Weightage**
1	Internal Assessment (PS): continuous evaluation of experiment performance, the portfolio/lab record, mini project and viva voce, and attendance	50 %
2	End Semester Practical Examination (PE)	50 %

Back to Course Structure & Evaluation Scheme

BT Stream

Course Code	**AAC101 / AAC201**
Title of the Course	Environment & Sustainability
Course Category	Mandatory Non-Credit / Audit Course (AU)
Teaching Load (L-T-P)	2 – 0 – 0
Credits	00 (Non-Credit)

Course Objectives:

1. To create awareness about the environment, natural resources, and ecological balance
2. To understand the causes, effects, and control of environmental pollution
3. To understand the concept of sustainable development and the UN Sustainable Development Goals (SDGs)
4. To understand climate change, renewable energy, and the engineer's role in sustainability
5. To encourage students to adopt simple sustainable practices and environmental responsibility

Course Content

Unit	**Description**	**Hours**
1.	**Environment and Ecosystems:** Environment — definition, components (atmosphere, hydrosphere, lithosphere, biosphere); ecosystem — structure and function; food chain, food web, energy flow (basic concept); biodiversity — meaning, importance, and threats; need for conservation. Awareness focus: local ecosystems and why they matter.	6 hrs
2.	**Natural Resources and their Conservation:** Renewable and non-renewable resources; water resources — scarcity and conservation; forest and land resources; energy resources (brief); over-exploitation of resources; the concept of the 3 R's — Reduce, Reuse, Recycle; rainwater harvesting (concept). Awareness focus: simple resource-saving practices in daily life.	6 hrs
3.	**Environmental Pollution:** Air pollution — causes, effects, control (basic); water pollution — sources, effects, prevention; soil pollution; noise pollution; solid waste management — segregation, composting, recycling; e-waste and plastic waste (awareness). Awareness focus:	6 hrs

BT Stream

	pollution in the student's own city/campus and how to reduce it.	
4.	**Climate Change and Sustainable Development:** Global warming and greenhouse effect (simple explanation); climate change — causes and impacts; carbon footprint (concept + how to reduce); sustainable development — definition and need; UN Sustainable Development Goals (SDGs) — overview of the 17 goals; renewable energy — solar, wind, biomass (awareness level). Awareness focus: calculating and reducing one's own carbon footprint.	6 hrs
5.	**Environmental Protection, Law and the Role of Engineers:** Environmental protection measures; brief overview of environmental laws in India (Environment Protection Act, Water Act, Air Act — awareness only, no legal detail); role of individuals, society, and technology in environmental protection; green campus and green buildings (concept); role of engineers in sustainable design and clean technology; environmental ethics. Awareness focus: how each engineering branch can contribute to sustainability.	6 hrs

Suggested Field / Awareness Activities (any one to be completed)

1. Visit a nearby lake, river, park, or forest and prepare a short observation report
2. Conduct a waste segregation / plastic-free drive on campus and document it
3. Prepare a poster or short video on an environmental issue (water, air, plastic, climate)
4. Calculate your household or personal carbon footprint and suggest ways to reduce it
5. Plant and maintain a sapling; document its growth over the semester (Green initiative)
6. Survey energy or water use in a hostel/department and suggest saving measures

Course Outcomes (COs)

COs	Course Outcome Statement	KL
CO1	Explain the components of environment and ecosystems and the importance of biodiversity	K2
CO2	Describe natural resources and identify practical measures for their conservation	K2

BT Stream

CO3	Explain the causes and control of environmental pollution and waste	K2
CO4	Explain climate change, carbon footprint, and the UN Sustainable Development Goals	K2
CO5	Identify the role of engineers and individuals in environmental protection and sustainability	K2

Co-PO Articulation Matrix

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	1	–	–	1	1	3	1	–	–	–	1
CO2	2	2	1	–	1	2	3	1	–	–	–	1
CO3	2	2	1	1	1	2	3	2	–	–	–	1
CO4	2	2	1	1	1	2	3	2	–	–	–	2
CO5	2	2	2	–	1	3	3	3	1	1	–	2

Evaluation Pattern

Sr.	Component	Weightage
1.	Assignment / Quiz / Class Test (awareness-based), Attendance and class participation	30%
2.	End Semester Examination (ESE)	70%

Suggested Learning Resources

1. Textbook of Environmental Studies for Undergraduate Courses – Erach Bharucha (UGC recommended)
2. Environmental Studies – Anindita Basak (Pearson)
3. Environmental Science – Y.K. Singh
4. UN Sustainable Development Goals – sdgs.un.org (free, official)
5. NPTEL – Environmental Studies / Sustainability courses (free video lectures)
6. Ministry of Environment, Forest & Climate Change (MoEFCC), Govt. of India – moef.gov.in

Back to Course Structure & Evaluation Scheme

BT Stream

STANDARD OPERATING PROCEDURE (SoP)

Innovation & Design Thinking Lab (AID151/AID251) — Conduct Guidelines

Purpose: This SoP provides a standard, uniform procedure for conducting the Innovation & Design Thinking Lab across all departments, ensuring consistent delivery, fair assessment, and achievement of course outcomes in line with the AICTE Design Thinking framework.

BT Stream

1. Team Formation

1. Students are organised into teams of 4–5 members at the start of the semester.
2. Teams should be interdisciplinary/mixed wherever possible to encourage diverse thinking.
3. Each team elects a Team Lead (role rotates across project stages so every member leads once).
4. Each team maintains a shared Design Journal (physical logbook or digital document) from Day 1.

2. Weekly Session Structure (1 hr Tutorial + 2 hrs Studio)

1. Tutorial hour: instructor introduces the design-thinking concept/tool for the week with examples.
2. Studio hours (2 hrs): teams apply the tool to their own project under faculty facilitation.
3. The faculty acts as a facilitator/mentor, NOT a lecturer — guiding, questioning, and unblocking teams.
4. Every session ends with a 5-minute stand-up where each team states progress and next steps.
5. All work (notes, sketches, photos) is recorded in the team Design Journal the same day.

3. Stage-Gate Milestones (Aligned to 5 Design Thinking Stages)

Week	Design Thinking Stage	Deliverable / Gate
1–2	**Introduction + Team formation**	Team formed, problem area chosen, project charter
3–5	**Empathize**	User research, interviews, empathy map, personas

6–7	**Define**	Refined problem statement (POV) + How-Might-We questions
8–9	**Ideate**	Brainstorm record, shortlisted concept, concept sketch
10–12	**Prototype**	Low-cost working prototype / model built
13–14	**Test & Iterate**	User testing feedback + at least one iteration
15	**Pitch**	Final pitch, prototype demo, design journal submission

4. Role of Faculty Facilitator

1. Facilitate, do not dictate — let teams discover solutions themselves.
2. Ensure every team member participates; prevent free-riding through individual check-ins.
3. Provide timely feedback at each stage-gate before allowing progress to the next stage.
4. Arrange access to basic prototyping materials and tools (see Section 6).
5. Maintain a facilitation log recording each team's progress and individual contributions.
6. Invite an industry expert / entrepreneur / senior alumnus for the final pitch wherever possible.

5. Assessment Procedure

1. Internal Assessment (50 marks) is continuous and stage-wise, per the IA rubric in the syllabus.
2. Each stage-gate deliverable is assessed and recorded before the team proceeds.
3. Individual marks may vary within a team based on documented individual contribution.
4. External Assessment (50 marks) is conducted at semester end: final prototype demo + pitch + viva by an external examiner.
5. A team that has not cleared at least the Prototype stage-gate is not eligible for the final external assessment.
6. Design Journal is mandatory for both IA and EA — teams without a journal cannot be assessed.

6. Required Infrastructure & Materials

1. Design studio / open classroom with movable furniture for team work.
2. Whiteboards / chart paper, sticky notes (Post-its), markers, sketch pens.

BT Stream

3. Basic prototyping materials: cardboard, chart paper, glue, scissors, clay, thermocol, tape, craft materials.
4. Access to a computer lab with free tools — Tinkercad, Canva, Figma, Google Workspace.
5. Access (shared) to a 3D printer / maker-space / tinkering lab where available (ATL/IIC).
6. Projector / smart board for pitches and concept sharing.

7. Safety, Ethics & Academic Integrity

1. Students must handle tools (cutters, glue guns, 3D printers) safely and under supervision.
2. User research must be respectful; informed consent must be taken for interviews and photographs.
3. Projects must be original team work; plagiarism of existing products without acknowledgement is not permitted.
4. Teams are encouraged to consider social impact, inclusivity, and sustainability in their solutions.
5. Any promising innovation may be forwarded to the Institution's Innovation Council (IIC) / incubation cell for further support.

8. Documentation & Records to be Maintained

1. Team-wise Design Journal / Portfolio (empathy maps, sketches, photos, test notes).
2. Faculty facilitation log (per team, per session).
3. Stage-gate assessment sheets with marks.
4. Final project report (short) + pitch deck + prototype photographs.
5. Course file: syllabus, SOP, CO-PO mapping, sample deliverables, and result analysis.

Note: *This SOP is aligned with the AICTE model curriculum on Design Thinking & Innovation and the objectives of the Institution's Innovation Council (IIC) under the Ministry of Education's Innovation Cell. Departments may adapt studio scheduling to local timetable while retaining the five-stage design-thinking flow and stage-gate assessment.*

Back to Course Structure & Evaluation Scheme

BTStream

STANDARD OPERATING PROCEDURE (SoP) for General Proficiency

1. Title

Standard Operating Procedure (SoP) for implementation of **General Proficiency**

2. Objective

1. To promote **holistic development** of students beyond academic learning
2. To inculcate **discipline, ethical values, and social responsibility**
3. To enhance **communication, leadership, and teamwork skills**
4. To align student development with the vision of **NEP 2020**
5. To prepare students for **professional and societal challenges**

3. Scope

This SoP shall be applicable to:

- All **undergraduate (B.Tech) students**
- All **affiliated and constituent institutions** of the University
- All semesters (or as notified by the University)

4. Course Structure

Parameter	Details
Course Name	General Proficiency
Marks	50
Credits	Non-Credit
Nature	Mandatory
Type	Internal Assessment (Non-Theory)

5. Components of Assessment

Category	Marks	Key Evaluation Parameters
Discipline & Ethics	10	Attendance, punctuality, adherence to rules, respect for faculty and peers, ethical behavior
Communication & Leadership Skills	10	Language proficiency, confidence, presentation skills, leadership, problem-solving
Co-curricular Activities	10	Participation in debates, seminars, sports, cultural events, social service

BT Stream

Teamwork & Collaboration	10	Group participation, coordination, cooperation, helping attitude
Professional & Technical Skills	10	Practical knowledge, digital literacy, innovation, creativity

6. Evaluation Process

- Evaluation shall be conducted **semester-wise**
- Assessment shall be based on:
  - ✓ Continuous observation by faculty
  - ✓ Participation in activities
  - ✓ Performance in institutional and external events
- Students may be given opportunity for:
  - ✓ **Self-assessment**
  - ✓ **Faculty assessment**

7. Implementation Mechanism

- Each department shall nominate a **Faculty Coordinator**
- A **Proficiency Assessment Committee** may be constituted at institute level
- Records shall be maintained in:
  - ✓ Digital format
  - ✓ Activity log / portfolio of students

8. Documentation & Record Keeping

- Students shall maintain:
  - ✓ Participation certificates
  - ✓ Activity reports
  - ✓ Portfolio of achievements
- Institutes shall maintain:
  - ✓ Attendance and evaluation records
  - ✓ Consolidated marks sheet

9. Grading & Reporting

- Marks shall be awarded out of **50**

BT Stream

- The score shall be:
    - ✓ Reflected in **Grade Card**
    - ✓ Considered for **overall student evaluation (non-SGPA component, if applicable)**

10. Alignment with NEP 2020

The course aligns with:

- **Holistic Development (Para 4.4)**
- **Co-curricular Integration (Para 4.8)**
- **Ethics & Value-based Education (Para 4.27)**
- **Life Skills & Multidisciplinary Learning (Para 11.5)**
- **Student Well-being (Para 4.42)**

11. Expected Outcomes

Students will be able to:

- Demonstrate **discipline and ethical behavior**
- Exhibit **effective communication and leadership skills**
- Participate actively in **co-curricular and social activities**
- Work effectively in **teams**
- Apply **technical and professional skills in real-life situations**

12. Review and Monitoring

- Periodic review shall be conducted by:
    - ✓ **Departmental Committee**
    - ✓ **Academic Council (if required)**
- Feedback from students and faculty shall be considered for improvement

Back to Course Structure & Evaluation Scheme

BT Stream

STANDARD OPERATING PROCEDURE (SoP) for Academic Activity Points (AICTE Activity Point Programme)

1. Title

Standard Operating Procedure (SoP) for implementation of **Academic Activity Points for Award of Degree**

2. Objective

- To promote **holistic development** of students beyond academics
- To encourage **community engagement, innovation, and social responsibility**
- To develop **leadership, teamwork, and entrepreneurial skills**
- To align with **AICTE guidelines and NEP 2020**

3. Scope

This SoP shall be applicable to:

- All **B.Tech students**
- All **affiliated and constituent colleges**
- Students admitted under **regular, lateral entry, and transfer categories**

4. Activity Points Requirement

Student Category	Minimum Activity Points
Regular Students (4-Year Programme)	100
Lateral Entry Students	75
Transferred Students	50

5. Nature of Activity Points

- **Mandatory for award of degree**
- **Non-credit** (not counted in SGPA/CGPA)
- Reflected in **final semester grade card**

6. Activity Categories and Approved Activities

Activity Summary Table

Sl. No.	Activity Head	Minimum Duration (Weeks)	Hours	Maximum Points	Evaluated By

BT Stream

				per Activity	
1	Helping local schools to improve results and enhance enrolment	2	80–90	20	Coordinator NSS / Youth Red Cross/ College Internal Compliant Committee / Mentor
2	Preparing business proposal for village income enhancement	2	80–90	20	Same
3	Sustainable water management system development	2	80–90	20	Same
4	Tourism promotion initiatives	2	80–90	20	Same
5	Promotion of appropriate technologies	2	80–90	20	Same
6	Energy conservation initiatives	2	80–90	20	Same
7	Skill development for rural population	2	80–90	20	Same
8	Digital financial inclusion initiatives	2	80–90	20	Same
9	Women empowerment awareness programmes	2	80–90	20	Same
10	Waste management system development	2	80–90	20	Same
11	Marketing support for rural produce	2	80–90	20	Same
12	Food preservation and packaging	2	80–90	20	Same
13	Automation of local activities	2	80–90	20	Same

BT Stream

14	Rural awareness programmes	2	80–90	20	Same
15	Participation in national initiatives (Swachh Bharat etc.)	2	80–90	20	Same
16	Rainwater harvesting awareness programmes	2	80–90	20	Same

7. Activity Duration and Completion

- Minimum duration per activity: **2 weeks**
- Minimum engagement: **80–90 hours**
- Activities can be completed during:
    - ✓ Weekends
    - ✓ Vacations
    - ✓ Semester breaks

8. Evaluation Mechanism

- Evaluation shall be carried out by:
    - ✓ Faculty Mentor
    - ✓ NSS / Activity Coordinator
    - ✓ Institutional Committee
- Assessment criteria:
    - ✓ Participation
    - ✓ Quality of work
    - ✓ Impact and outcomes

9. Implementation Framework

- Each institute shall appoint:
    - ✓ **Activity Point Coordinator**
    - ✓ **Monitoring Committee**
- Responsibilities:
    - ✓ Approve activities

BT Stream

- ✓ Monitor student progress
- ✓ Validate submitted documents

10. Documentation Requirements

Students must maintain:

- Activity reports
- Certificates
- Proof of participation (photos/videos)
- Mentor verification

Institutes must maintain:

- Student-wise records
- Consolidated activity reports

11. Submission and Reporting

- Institutes shall submit **consolidated reports to the University**
- Activity points shall be verified before **final semester completion**
- Points shall be reflected in the **Grade Card**

12. Compliance Requirement

- Earning required Activity Points is **mandatory for award of degree**
- In case of non-compliance:
  - ✓ **Degree shall not be awarded**
  - ✓ Final Grade Card may be withheld

13. Monitoring and Review

- Periodic review at:
  - ✓ Department level
  - ✓ Institute level
- University may conduct:
  - ✓ Audits
  - ✓ Compliance checks

14. Expected Outcomes

BT Stream

Students will:

- Develop **social responsibility and ethical values**
- Gain **practical exposure and real-world experience**
- Improve **leadership and teamwork skills**
- Enhance **employability and innovation capability**

15. Alignment with NEP 2020

- Holistic and multidisciplinary education
- Skill-based and experiential learning
- Community engagement
- Value-based education

Back to Course Structure & Evaluation Scheme

BT Stream